FSM PLATFORM

SYSTEM MANUAL

# FSM User Manual

Office · Technician · Administrator

A practical guide for office staff, technicians and administrators.

| VERSION | DATE | AUDIENCE |
| --- | --- | --- |
| 0.8.9 | 2026-11-04 | Office · Technician · Administrator |

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# Introduction and system overview

*Applies to FSM Platform version 0.8.9, 4 November 2026.*

## What FSM Platform is

FSM Platform is a service and maintenance system for work carried out at the customer's site: service, maintenance, inspection and repair. It brings planning, work orders, visits, checklists, reports, equipment, service contracts, documents and the customer portal together in one service.

The platform is designed so that the office can plan without parallel spreadsheets, the technician can work from a phone or tablet even with a poor connection, and the customer can see contracts, visits, reports and equipment without calling the office. The system documents and plans work; it does not record hours, calculate prices or issue invoices. Price lists and contract amounts are reference data.

### The four roles

This manual uses four standard roles. An administrator can create custom roles and give individual users tailored permissions, but day-to-day work generally follows these four.

| Role | Who | Main task | They see |
|---|---|---|---|
| **Administrator** | System or operations manager | Sets up departments, roles, templates, branding, API access and security | Everything, across all departments |
| **Office** | Service manager, planner or case handler | Plans and dispatches work, follows up customers, contracts, bookings and reports | Own department for work orders, contracts and reports; customers, equipment and documents are shared |
| **Technician** | Field service technician | Carries out visits, completes checklists, takes photos and signs | Own department for work orders and reports; customers, equipment and documents are available |
| **Customer portal user** | Customer or contact | Books service, follows visits, reads reports and views their own equipment | Their own customer, with the portal access granted |

> **Note:** "Office" is a role, not a physical place. A service manager can work from a phone and still have the office role.

## What's new in 0.8.9

- **Two-way SMS, office inbox and reminders.** Incoming SMS lands in the same conversation as outgoing messages, with delivery status and STOP/START consent. The office reads and replies under **Messages**, and an administrator can configure automatic SMS/email reminders with quiet hours and a per-visit cap. See the office chapter and the administrator chapter.

- **SSO and SCIM.** OIDC sign-in with Microsoft Entra ID, Keycloak, Authentik or Zitadel, plus SCIM 2.0 user provisioning. Password sign-in remains the break-glass path. See the administrator chapter.
- **Field app camera and QR.** Scanning works on iOS and Android over HTTPS, using the browser barcode API where available and a fallback elsewhere. Manual codes, navigation and existing QR labels from other systems are supported. See the technician chapter.
- **PDF in the app or as a download.** Reports and revisions can be shown safely in the app's PDF viewer after sanitisation. Direct links and **Download PDF** still produce a file. See the office chapter.
- **Customer transfer.** An administrator can export and import customer packages as ZIP, with preview, checksums and a single import transaction. See the office chapter.
- **Route planning and live ETA.** The route planner suggests driving order, drive time, distance and waiting time, with optional OSRM and a local fallback. The tracking page states how fresh the position is. See the office chapter and the administrator chapter.
- **Customer scoping and equipment moves.** Site and unit pickers load data only for the selected customer. A unit can be moved to another customer with its service history, after a preview and contract checks. See the office chapter.
- **Upload limits and quotas.** 15 MB per image, 50 MB per PDF, 2 GiB per user per day and 10 GiB per customer. The limits can be adjusted in the environment. See the administrator chapter.

## How the system is put together

You meet the system through a few surfaces. They share data and permission checks but are adapted to different tasks.

| Surface | What you do there |
|---|---|
| **Browser and PWA** | Opens the system on a PC, tablet or phone. The field surface can be installed as an app and used with limited connectivity. |
| **Office surface** | Plans in lists, boards, calendars and dispatch. Handles customers, contracts, bookings, reports and settings. |
| **Field surface** | Shows today's jobs, the next action, map links, checklists, photos and signatures, with large touch targets and a dark theme. |
| **Customer portal** | Gives the customer an overview of their own bookings, reports, documents, equipment, contracts, visits and messages. |
| **REST API** | Lets other systems read and write data with a dedicated API key and the same permission checks. |

Behind the surfaces is a shared application that enforces workflow and access. It is built with Next.js and React, using Prisma against PostgreSQL as the source of truth. The field app is a PWA with a service worker and IndexedDB for offline drafts. PDFs and uploaded files live in a separate file volume. Report PDFs render in-process with a concurrency limit; OCR and document jobs run through a job queue.

| Layer | Content |
|---|---|
| **User surfaces** | Customer portal, office surface, field surface and REST API clients in a browser or installed PWA. |

| Layer | Content |
|---|---|
| **Application** | Access control, workflow, notifications, search and PDF rendering. |
| **Data and jobs** | PostgreSQL with data, revisions and audit log; file volume for PDFs and images; job queue for OCR, document jobs and notifications; email/SMS where configured. |

The field surface also has a local buffer in the browser:

1. **Offline:** the form is stored in IndexedDB as a draft and outbox item.
2. **Online:** the draft is sent to the server, which confirms receipt.
3. **Always:** the server decides whether a report has actually been received; "sent" appears only after confirmation.

A draft can be saved locally, but a submitted report is not considered sent until the server confirms it. This is why the field surface shows "draft saved locally" and "waiting to send" as two different states.

## Core entities

| Entity | What it is | Key relationships |
|---|---|---|
| **Department** | Organisational unit that determines who sees what | Staff, work orders, customers, contracts, business hours |
| **User and role** | Employee or customer portal user with permissions | Department, customer, contact, API keys |
| **Customer** | The company receiving service | Contacts, sites, equipment, contracts, work orders |
| **Contact** | The person the office and technician deal with | Customer, work orders, contracts, portal user |
| **Site** | Physical address or facility at the customer | Customer, work orders, equipment, bookings |
| **Equipment/unit** | Machine or system being serviced | Customer, site, meter readings, maintenance plans, history |
| **Work order** | The service job itself, with status and responsibility | Customer, contact, site, equipment, templates, visits, reports |
| **Visit** | Time block within a work order, with technicians and status | Work order, technicians, tracking, meter readings |
| **Checklist template** | Reusable template for service or HMS | Work orders, contracts, reports, maintenance plans |
| **Report, unit and answers** | Completed checklist with units, photos and signatures | Work order, template, equipment, PDF, revisions, acknowledgement |
| **Service contract and SLA** | Agreement for recurring service and deadlines | Customer, department, template, SLA policy, work orders |

| Entity | What it is | Key relationships |
|---|---|---|
| **Booking** | Customer's request or confirmed appointment | Customer, site, equipment, work order, visit, events |
| **Document** | File in the library or an attachment | Folder, tags, versions, work order, customer, contract |
| **Notification** | Message to a user about an event | User, type, source, link, email status |
| **Audit log** | Traceable history of actions | Entity, actor, time, metadata |
| **API key** | Secret for integrations | User, permissions, expiry, revocation |

## End-to-end flows

The role chapters describe each step in detail. Here you can see how the roles and data fit together.

### 1. Service contract → recurring work order → report → PDF and history

1. **Office or administrator** creates a service contract on the customer card with an interval (3, 6, 12 or 24 months), start and end dates, department, template, optional SLA and renewal mode.
2. **The system** creates a new work order when `Next service` is due. The work order inherits the customer, contact, template and department; the previous report is attached as a PDF.
3. **The office** sees the work order in "Work orders" and "Calendar", creates or checks the visit and assigns a technician.
4. **The technician** completes the checklist per unit, takes photos, comments on deviations and signs if desired. The customer can sign at the same time.
5. **The system** locks the report, saves a revision, generates a PDF and notifies the office and customer portal. If there are deviations, it creates one undated remediation work order.
6. **The customer portal user** sees the submitted report, downloads the PDF and acknowledges receipt. The acknowledgement is visible to the office.
7. **The office** follows up deviations. The equipment on the customer card shows service history from the report's units.

> **Note:** The report is frozen at submission. If the template or equipment changes later, the submitted report keeps the information that applied when it was submitted.

### 2. Customer portal booking → work order and notification → dispatch → technician

1. **The customer portal user** selects a site, service and requested date under "Bookings". The desired time is written in the description; the customer cannot select a time block.
2. **The system** creates a new work order with status `OPEN` and an unconfirmed booking request. No visit or confirmed time is created, and the office is notified.
3. **The office or technician** plans the work order and sets the status to **Planned**, or **Waiting** if the time does not fit. Only then are the date and time confirmed.

4. **The technician** receives an assignment notification and sees the visit under "Today". Cancellation triggers a new notification.

> **Tip:** A customer booking is a request, not a confirmed appointment. The office must plan it before the customer gets a final time.

### 3. Office creates a work order → calendar, board and assignment → field

1. **The office** selects "New work order" and enters a title, customer, contact, site, date, template and optionally tags and technician.
2. **The system** creates the work order and a visit in the same transaction when a date is set. New work orders start as `PENDING`; undated work orders have no visit.
3. **The office** distributes the work through "Assign technician", the calendar, the dashboard board or dispatch with suggestions and automatic assignment.
4. **The technician** sees the work order under "Today" or "Work orders" and can claim available work with "Take work order" if the role allows it.
5. **The technician** starts driving, marks "On site", completes the form and finishes. Status, comments and the report flow back to the office.
6. **The office** follows up overdue work orders, deviations, missing signatures and reports that have not been acknowledged.

> **Note:** On the board, a drop assigns a technician; it does not move the date. Date and time are changed in the calendar or in the visit details.

### 4. Tracking and reviews → customer portal and office follow-up

1. **The administrator** enables live tracking for the department. It is off by default and applies only to visits in that department.
2. **The technician** sets the visit to "On the way" and can share position. The customer opens the tracking link from "Visits" or receives it separately.
3. **The customer portal user** sees the first name, status and estimated arrival. The position is never shown as a map or coordinates, and the link expires when the visit is finished or after two hours.
4. **The customer** can leave an internal review after the visit is completed. Reviews are internal and never published; they are not shown as a separate dashboard in 0.8.9.

### 5. Permissions, notifications and audit across all flows

1. **Every** action is authorised on the server. Staff see their own work orders, contracts and reports, while customers, equipment and documents are shared. Administrators see everything, and customer portal users see only their own customer and granted portal areas.
2. **The system** notifies on assignment, date change, cancellation, completed report, booking, contract due, overdue work orders, SLA events and contact requests. Notifications go to the bell, "Notifications" and optionally email.
3. **The audit log** records who created, changed, moved, cancelled, deleted, unlocked or applied a legal hold, with time and actor. The log cannot be edited.

4. **The administrator** can change roles and permissions. Old sessions are rejected, and API keys inherit the user's current permissions.
5. **Legal hold** prevents deletion and changes. A submitted report can only be unlocked by an administrator, with a reason, and the unlock is logged.

## Navigation by role

The menu is adapted to role and permissions, so you will not necessarily see every item.

| Role | Home | Main menu | Hamburger menu and settings |
|---|---|---|---|
| **Administrator** | "Dashboard" | Dashboard, Work orders, Calendar, Customers, Messages, Reports | Bookings, Contracts, Units, Search, HMS, Documents, Map, New work order, Notifications, Settings |
| **Office** | "Dashboard" | Dashboard, Work orders, Calendar, Customers, Messages, Reports | Bookings, Contracts, Units, Search, HMS, Documents, Map, New work order, Notifications, Settings |
| **Technician** | "Today" | Today, Work orders, Calendar, Customers, Messages | Reports, Bookings, Contracts, Units, Search, HMS, Documents, Map, New work order, Notifications, Customers & equipment, Settings |
| **Customer portal user** | "Customer portal" | Overview, Bookings, Reports, Documents, Equipment, Contracts, Visits, Messages, Contact | Account & security |

On mobile, the bottom bar shows the most important areas: "Dashboard", "Work orders" and "Calendar" for office and administrator, and "Today", "Work orders" and "Notifications" for technicians. "More" opens the rest.

### Where do I find …?

| What you are looking for | Go to |
|---|---|
| Today's work and overdue work orders | "Dashboard" for the office, "Today" for the technician |
| Planning or moving a visit | "Calendar" or the work order's visit section |
| Assigning technicians quickly | "Dashboard" → board, "Calendar" → Dispatch, or "Assign technician" |
| Customers, contacts and portal users | "Customers" → customer card |
| Equipment and service history | "Units" or the customer card → "Equipment" |
| Service contracts and next service | "Contracts" or the customer card → "Contracts" |
| Bookings from the customer portal | "Bookings" |
| Reports and PDFs | "Reports", or the work order → "Checklists" |

| What you are looking for | Go to |
|---|---|
| Checklists and HMS templates | "Settings" → "Checklists / Templates", or "HMS" |
| Documents, folders and tags | "Documents" |
| Notifications and notification settings | The bell at the top, or "Notifications" → "Settings" |
| Audit log for one work order | The work order → "Activity log" |
| Global audit log | "Settings" → "Activity log" (administrator) |
| Employees, roles and permissions | "Settings" → "Employees & Roles" or "Roles & permissions" |
| API keys | "Settings" → "API keys" (administrator) |
| Anything across customers and work orders | "Search" |

## How to use this manual

The manual has an introduction, three role chapters and a reference section. You do not need to read everything in order. Start with your role, and use the reference section when you need a term, a status or a troubleshooting check.

- **Role chapters:** Part 1 for office, Part 2 for technician and Part 3 for administrator. The customer portal is described alongside the customer card and the roles that administer it.
- **Reference:** the reference part contains the status glossary, permission matrix, terms and troubleshooting.

### Conventions

- Buttons, tabs, menu items and field names appear in "quotes", as they appear in the interface.
- Paths, file names, codes and technical values appear in `monospace`.
- Menu paths are shown as **Settings** → **API keys**.
- **Note** explains something that is easy to misunderstand. **Tip** is a shortcut or good practice. **Warning** means the action can lead to data loss or errors.
- Dates and times are shown in Europe/Oslo with a 24-hour clock. "Today" means the day in that time zone.
- When a time or date is missing, the system shows "Time not set" or "No date". That does not mean the value is midnight or that the job has been forgotten.
- Service report PDFs use Norwegian labels in 0.8.9, even when the interface is in English. Contract PDFs can be generated in the selected language.

> **Warning:** A submitted and locked report should not be changed without a good reason. Unlocking requires an administrator and a reason, and every change is recorded in the audit log.

On first sign-in you arrive at your role's home surface. Check the department, notification settings and any 2FA. Technicians should install the field surface as an app, so drafts and recent data are available even with an unstable connection.

PART 1

# Office: planning, dispatch and follow-up

This chapter is for office staff: you receive enquiries, create and plan work orders, assign technicians, follow up reports and keep customer, equipment and agreement data up to date. It applies to FSM Platform 0.8.9 (4 November 2026).

Work orders and service agreements belong to a department: you normally see only your own department, while an administrator sees everything. Customers, equipment and documents are shared, but every action requires the right permission. The platform does not issue invoices and does not record hours or materials. Price lists are reference data; invoicing happens outside the platform.

## Dashboard

The dashboard is at `/` and is titled **Planning**.

### List and board

**List** (`/?vis=liste`) is the default and shows one prioritised queue with **Overdue**, **Today** and **Unassigned**.

**Board** (`/?vis=tavle`) has the columns **Overdue**, **Today**, **In progress** and **Unassigned**. These are priority buckets, not statuses; the status is a pill on the card.

| Card | Meaning | Click |
|---|---|---|
| **Overdue** | Scheduled before today, not completed | `/oppdrag?vis=forfalte` |
| **Today** | Scheduled today, including multi-day work orders | `/oppdrag?vis=idag` |
| **Unassigned** | Active work order without a technician | `/oppdrag?vis=ikke-tildelt` |
| **In progress** | Status **In progress** | `/oppdrag?vis=pagaaende` |

The **Queue** can be sorted by **Date**, **Customer**, **Status** or **Title**; the default is **Oldest first**. The right-hand rail shows **Next action** with **Assign technician** or **Open work order**, **Today's progress** and **Team today**.

### Exceptions, quick actions and activity

At the top of the dashboard, **Exceptions needing action** shows the cards **Unassigned today**, **Visits without technician**, **Double booked**, **Overdue**, **Missing coordinates**, **Unsigned reports** and **Stale bookings**. Each card shows a count and opens the relevant list or dispatch. **Quick actions** gives **New work order**, **Calendar**, **Dispatch**, **Search**, **My work orders**, **All work orders**, **Customers**, **Documents**, **Checklists**, **Statistics** and **Map**. **Recent activity** shows the latest events with **View all activity**.

### Quick assignment

1. Go to **Work orders** (`/oppdrag`).
2. Find the row that says **No technician assigned**.

3. Choose a technician in **Choose technician** and press **Assign**.

The technician appears in the row. The assignment applies to the work order; if only one visit needs a different technician, do it in **Planning**.

### Drag and drop

Drag a card from the board to a technician in **Team today**, or from **Resource overview** to the technician's column. Drag and drop assigns a technician, **not** a date; use the calendar or **Planning** to move time.

### Global search

**Search** in the top bar searches customers, work orders, documents, units and reports. Results show their type, and the search tolerates typos. Document hits open the document in the library. Use search when you only have a name, an order number or a keyword.

## Work orders

The work order list is at `/oppdrag` with the views **Active**, **New**, **Not started**, **In progress**, **Today**, **Overdue**, **Unassigned**, **Completed** and **Cancelled**.

### Create

1. Press **New work order** ( `/oppdrag/ny` ). The department is fixed to your own; only an administrator can choose another. Enter **Work order title** and optionally **Order number** (unique in the department).
2. Enter **Description / Instructions**.
3. Choose **Customer** and **Contact person**, or leave the customer empty for an **Internal work order**. **Address**, **Postal code** and **City** override the customer address.
4. Set **Date of execution** and optionally **End date (optional)**.
5. Choose technicians under **Assign technicians** and templates under **Add checklists / forms**, and add **Tags** if needed.
6. Press **Save work order**. You return to **Calendar**.

**Edit work order** changes title, customer, contact, date, technicians and description.

### Priority and status

There is no separate priority field; priority comes from the date, the status and whether a technician is missing. The statuses are **Open / New**, **Waiting to be planned**, **Planned**, **In progress**, **Waiting for customer**, **Completed** and **Cancelled**. Only an administrator can reopen a completed or cancelled work order.

### Checklists, technicians and visits

On the create page, templates are grouped under **Service / checklists** and **HMS documentation**. Each template produces its own report, which is always new and empty.

You can assign several technicians; **Remove** takes one off. The work order has a date; the time is set on the visit:

1. Open **Planning** and press **Add visit**.
2. Choose **From date**, optionally **Through**, and enter **Start time** and **End time** in 24-hour format, for example `09:00` and `11:30`.
3. Tick **All day** if the job has no fixed time, choose **Technicians on the visit**, and add a **Note** if needed.
4. Press **Save visit**.

A visit lasts at most 31 days, and the end time must be after the start time. Overlapping visits are allowed. The technician is shown as busy, but can still be assigned another job; use the calendar and capacity view to judge the load.

## HMS library and QR labels

**HMS** (`/hms`) collects HMS forms by type: **Safe Job Analysis (SJA)**, **Hot work**, **Internal control**, **Deviations**, **Safety round**, **Written instruction** and **General**. Under **Link HMS form to work order**, choose an **HMS form** and an active **Work order**, then select **Link form to work order**. The form is added to the work order and filled in there. **Recent HMS documentation** shows completed HMS forms you have access to.

QR label: open the unit at `/enheter/<id>`, select **Print label**, and use **Print** on the label page (`/enheter/<id>/etikett`). Scanning the label opens the unit. **Rotate QR token** if the label is lost or shared; the old label stops working immediately.

**Existing QR codes:** If the customer already has labels from another system, register the value under **Existing QR codes** on the asset page. Paste or scan the value, optionally give it a name (for example “Old QR – gate 3”), and save. The label stays physically on the equipment and is never changed. Each value can be linked to only one asset. Deleting removes only the mapping in FSM, not the physical label, and is recorded in the audit log.

## Attachments, comments and completion

**Attachments** accepts images and PDFs, up to 15 MB per image and 50 MB per PDF, and is internal. A rolling 24-hour quota of 2 GiB per user and 10 GiB total storage per customer also applies; adjust it with `UPLOAD_QUOTA_USER_MB` and `UPLOAD_QUOTA_CUSTOMER_MB`. **Comments** are the dialogue between office and technician; write in the field and press **Add comment**.

**Complete** sets the status to **Completed** after the confirmation **Complete this work order? The work order and report are locked for editing. Cancel work order** requires a **Reason for cancellation**; the customer is notified, and only an administrator can reopen it.

## Clone, related data and trash

**Clone work order** copies the details, customer, equipment, technicians, templates, tags and visits. The clone gets the status **Waiting to be planned**, and the **order number is cleared**. Reports and attachments are not copied.

| Data | Where | Note |
|---|---|---|
| Customer/contact | Header and **Location and contact** | Locked when a report or booking exists |
| Equipment | Customer card → **Equipment** | The report remembers the units checked |

| Data | Where | Note |
|---|---|---|
| Reports | **Checklists and reports** | Locked when submitted |
| Visits | **Planning** | Control the date period |
| Agreement/booking | The agreement's latest work order / booking queue | One booking per work order |

**Move to trash** is administrator-only by default; the button is shown only to roles with `workorders.delete` and only for work orders that are not closed. An administrator can restore it from **Trash** (`/oppdrag/slettede`). Work orders without reports can be removed after 365 days; work orders with reports are retained. **Legal hold** stops all automatic deletion.

# Customers and contacts

The customer directory is at `/kunder`. Search by name, org. number, city or email, or press **New customer**.

The customer card has the tabs **Overview**, **Work orders**, **Agreements**, **Equipment** and **More**. At the top you find **New work order**, **Messages**, **Edit** and, for an administrator, **Delete customer**. **Overview** shows company information, contacts, portal users, overdue and upcoming work orders.

## Contacts

1. Open the customer card → **More** → **Contacts**.
2. Press **Add contact**.
3. Enter **Name**, **Role (optional)**, **Email (optional)** and **Phone (optional)**, and save.

A portal user must always be linked to a contact. If you delete the contact, any portal users linked to it are deactivated.

## Department and responsible technician

**Edit customer** has **Department (optional)** and **Responsible technician (optional)**. The responsible technician is shown as the customer's direct contact.

## Portal users

Customer card → **More** → **Portal users**.

1. Press **Create and send invitation**.
2. Enter **Name**, **Email** and choose **Contact**.
3. Send the invitation; the user sets a password via the link.
4. Press **Activate** once the password is set.

If the email cannot be sent, the system shows **Invitation ready for {name}** with **Copy link**; send it manually. **Deactivate** closes the portal and signs the user out.

### Equipment, documents and agreements

**Equipment** shows units and service history. **More** → **Attachments** holds documents, agreements and images belonging to the customer. **Agreements** shows service agreements, coverage lines, next service, SLA and price lists.

**Sites** and **Equipment** on the customer card have their own search fields. The search runs on the server and is limited to the open customer, so you can search by name, address, postal code, city, model or serial number without other customers' rows appearing. **Reset** clears the search, and the counter shows how many matches the customer has.

## Equipment and assets

The asset register is at `/enheter`. Search by model, serial number, location or customer, filter by status and category, or turn on **Show hierarchy**.

### Create and edit

1. Press **New asset** (`/enheter/ny`).
2. Choose **Customer**, and enter **Model** and **Serial number**.
3. Enter **Location**, **Category**, optional **Parent asset**, **Status** and **Criticality**.
4. Set **Installation date**, **Warranty until** and **Meter unit** if relevant.
5. Press **Register asset**.

Once a **Customer** is selected, only that customer's **Sites** and **Parent asset** are loaded from the server. Changing customer clears and reloads the lists, so you cannot select a site or asset belonging to another customer. The same applies to **New work order** and **Edit work order**.

**Edit asset** changes the same fields.

The statuses are **Active**, **Out of service** and **Retired**; criticality is **Critical**, **Normal** or **Low**. **Parent asset** builds a hierarchy, and **Move in hierarchy** changes placement.

### Move a unit to another customer

When equipment changes owner or customer, the whole unit can be moved with its service history. The move requires **Manage equipment** and **Manage customers**, or an administrator.

1. Open the unit at `/enheter/<id>` and select **Move to another customer**.
2. Search for the target customer and optionally choose a target site. If you choose no site, the unit stays without a site at the new customer.
3. Review **What moves** and the preview. It shows the number of work orders, visits, reports, revisions, readings, plans, documents, attachments and QR codes that will follow.
4. Type the target customer's name to confirm, or tick the confirmation, and select **Move the unit**.

**This moves:** the unit and any child units (up to 1000), the hierarchy between them, work orders and visits, reports with revisions, report units and PDFs, readings, maintenance plans and occurrences, documents with versions, attachments/uploads, registered QR codes and comments. The root unit receives the target site. Child units lose the old site because it belonged to the previous customer and must be placed again after the move. The root unit is no longer linked to any former parent unit.

**This blocks the move:**

- **Contract lines belonging to another customer's agreement** that point to the unit or a child unit. A move cannot silently change who a financial obligation points to. Clear or end the contract lines first.
- Child units that already belong to another customer.
- Order numbers that collide with a number in the target customer's department.
- A missing or deleted target customer, or a target site that does not belong to the customer.
- More than 1000 units in the subtree.

**This does not move:** contract lines, the old customer's portal users, global documents without a customer or unit link, or the audit log itself. Contacts from the old customer are cleared on the work orders, and work orders that point to another customer's agreement lose the contract link. The preview warns you in advance. The target customer's department is used when it exists; otherwise your own department is used, and you are told.

The move is idempotent: repeating it after the unit is at the target customer reports that nothing changed. All validation runs before the first write, so a rejected move leaves no half-finished changes. The action is audit-logged. To undo it, move the unit back manually.

## Readings

1. Open the asset and find **Meter**.
2. Press **Register reading**.
3. Enter **Value**, **Date**, **Source** and optionally **Note**.
4. Press **Save reading**.

## Maintenance plans

1. Press **New maintenance plan**.
2. Enter **Title** and choose **Trigger**: **Calendar** or **Meter**.
3. For calendar: set **Interval (months)** and optionally **Lead time (days)**. For meter: set **Meter interval**.
4. Check **Next due** and save.

**Pause** stops the plan, and **Activate** starts it again; **Timeline** shows occurrences. When the plan falls due, the scheduled job creates a work order for the office to follow up.

## Service history and reports

**Service history** shows every report in which the asset was checked, with date, result, comments, checkpoints and images.

In the form, the technician chooses **Get from the customer equipment**. Each registered asset can be used only once in the same report. The report remembers the asset, serial number and answers, so the customer card and asset page show exactly what was checked.

# Calendar and dispatch

The calendar is at `/kalender` with the tabs **Resource overview**, **Month**, **Week**, **2 weeks**, **Week (time)**, **Day**, **Dispatch**, **Capacity**, **Route** and **Reports**.

| Tab | URL |
|---|---|
| **Month** | `/kalender?vis=maned` |
| **Week** | `/kalender?vis=uke` |
| **2 weeks** | `/kalender?vis=2uker` |
| **Week (time)** | `/kalender?vis=timegrid` |
| **Day** | `/kalender?vis=dag` |

At the top, **Period** shows the month or date range right next to the navigation arrows and **Today**. The calendar loads about six months back and forward, and a maximum of 1000 visits; visits outside the window may be missing.

**Filter the calendar** has **Technician**, **Status** and **No technician**. **Apply filter** and **Reset** apply to all date views.

## Visits and drag and drop

The calendar shows visits with the status **Planned**, **Assigned**, **On the way**, **On site**, **Completed** or **Cancelled**. Drag a visit to another day or time range to move it. On conflict the move is not saved, and you are told about the overlapping assignment.

**Resource overview** shows technicians, active work, unassigned work and recent completions. **Unassigned work orders** have **Assign technician**. You can drag cards between technicians to assign.

## Dispatch and capacity

**Dispatch** ( `/kalender/dispatch` ) assigns technicians to visits, with a separate **No technician** column. **Overdue / Unassigned** is always shown, regardless of the date. Drag a visit to a technician's column to move it. Drop it on **No technician** to remove the source technician; other technicians are kept.

**Automatic assignment** is dry-run with **Dry run**; **Use {count} suggestions** saves only when you choose it. **Suggest technician** ranks candidates by **Skills**, **Distance**, **Travel**, **Free**, **SLA** and **Load**.

**Capacity** ( `/kalender/kapasitet` ) shows **Planned capacity against actual load per department, day and technician**. Choose **From** and **To**, press **Show**, and read **Capacity**, **Load** and **Utilisation**.

## Route plan

**Route** ( `/kalender/rute` ) suggests a driving order for one day, per technician. Choose **Date** and optionally **Technician**, then **Show route**. The page shows a map, stop numbers, **Arrival**, **Planned**, **Driving**, **Waiting**, and total **Drive time**, **Distance** and **Waiting time** per technician.

The route starts at the technician's earliest located visit. Visits without coordinates, or that do not fit within the time window, are listed under **Visits not included in the route** and are not moved. The calculation uses an

external routing provider when configured; otherwise straight-line distance × 1.35 and average speed. **Calculation** shows which method was used.

**Use order** moves the visits to the suggested times in one transaction. The action requires `workorders.assign`, is audit-logged per visit, and is cancelled entirely on overlap with a fixed visit or registered holiday/sickness. Completed and cancelled visits are never moved. If the plan has changed since it was shown, you are asked to reload the page.

**Assignment rules** and **Territories** are configured under **Settings** → **Dispatch**. There you choose a **Mode**, adjust the **Weights** for **Skills**, **Distance**, **Travel time**, **Free time**, **SLA** and **Load**, and add territories with **Postal prefixes**, **Municipality codes** and **Radius (km)**. **Capacity plans** are shown in the capacity view but are not edited in **Settings** in 0.8.9. The same applies to skills, absence and working hours: dispatch and booking use them, but they have no separate form in **Settings**.

# Agreements, SLA and price lists

Agreements are at `/avtaler` and on the customer card's **Agreements** tab.

## Create

1. Open the customer card → **Agreements**.
2. Enter **Title**, optional **Agreement number**, **Department**, **Start date** and **End date**.
3. Choose **Interval** (3, 6, 12 or 24 months), **Status** (**Draft** or **Active**) and **Renewal** (**Manual** or **Automatic**).
4. Set **Notice before expiry (days)** and optionally **Price adjustment (%)**.
5. Choose **Contact**, **Checklist**, **SLA policy** and **Price list**, and add a line with **Equipment**, **Coverage**, **Visits/year**, **Response (min)** and **Resolution (min)**.
6. Press **Create service agreement**.

## Lines and coverage

**Coverage ({count})** shows the agreement lines. An agreement **without lines** covers all equipment; a line **without equipment** covers all equipment; a line **with equipment** covers only that unit. Read equipment, coverage, visits per year and response times. Only **Active** agreements within the start and end dates count. No automatic “covered” badge is shown, so check manually before promising cover.

## Interval, renewal and lifecycle

**Next service** is calculated from the start date and interval. **Renew** extends; **Pause** and **Activate** stop and start new work orders. **Automatic** renewal is run by the scheduled service agreement job. **Manual** renewal becomes **Expired** after the end date.

## SLA

An SLA policy sets **Response time (minutes)** and **Resolution time (minutes)**. It can be department-specific or global, and limited to **Count business hours only**. **Business hours** and **Holidays** decide when the clock runs; without saved business hours, weekdays 08–16 are used. Policies, business hours, holidays and price lists are maintained by an administrator under **Settings** → **SLA and business hours**.

An agreement can point to a policy; otherwise the department default is used. On the work order, **SLA** shows **Respond by**, **Resolve by** and **On track**, **At risk**, **Breached**, **Met** or **Cancelled**. **On the way** or **On site** records response; **Completed** records resolution.

### Price lists

**Price lists are reference data for display. The system does not invoice or calculate amounts.**

### Recurring work orders

At each interval a new **Planned** work order is created with the agreement's customer, contact, department and checklist, plus a four-hour default visit without technicians (08:00–12:00). Expired agreements do not generate new work orders.

## Bookings

The booking queue is at `/bookinger` with **All**, **Pending**, **Confirmed** and **Closed**. Each card shows **Reference**, **Customer**, **Requested time**, **Location**, **Service**, **Technician** and status.

1. Open `/bookinger` and filter on **Pending**.
2. Choose **Technician** in **Choose technician**.
3. Press **Confirm booking**, or **Cancel** to decline.

On confirmation the work order becomes **Planned**, the visit is created or updated, and the customer receives an SMS if the contact has consented.

The customer chooses a site, service and requested date in the portal, and writes the desired time in the description. The customer cannot select a time block, and the booking always creates a new work order with status **Open/New**. No visit or confirmed time is created.

The office or technician plans the work order and sets the status to **Planned**. If the time does not fit, it is set to **Waiting** for replanning. The customer receives the final confirmation only then. A customer booking is already a work order; there is no separate **Convert to work order** button.

## Reports

The report list is at `/rapporter` and shows **Service reports** – completed checklists with signatures and photos.

### List and filters

Choose **This month**, **Last month** or **All**, and filter on **Free text**, **Customer**, **Technician**, **Deviations only** and **Acknowledgement** (**All**, **Acknowledged**, **Not acknowledged**). **Sorting** is **Newest first**, **Oldest first**, **Customer (A–Z)** or **Technician (A–Z)**. Press **Filter** or **Reset filters**. The list shows **Locked**, **Open for correction** or **Unsigned**, and **Report no.** when numbered.

### Open, PDF and content

**Open form** opens the report with **Main result**, **Signed by**, **Units that were checked**, **Answers to checkpoints** and **General notes from the technician**. The PDF is shown in the app's own PDF viewer.

**Open in new tab** shows the same PDF in the browser, **Download PDF** creates an attachment, and **Close PDF** goes back. If the PDF cannot be shown, for example because it contains active content that cannot be removed, the viewer shows an error with **Try again** and download. Revisions are opened the same way from **Revisions** using **Open revision (PDF)**.

A report can have a **Technician** and a **Customer** signature. Photos are added per checkpoint or unit. Deviations appear as **Deviation found**. When a report contains at least one deviation, the system automatically creates **one undated remediation work order** titled “Utbedring av avvik – …”. It inherits customer, department and contact, gets the submitter as technician, and describes the units and checkpoints. The generated title is Norwegian (“Utbedring av avvik – …”) in 0.8.9, even when the interface is English.

> **Note:** If a photo or signature is missing from the finished PDF, the PDF shows a warning with the number of images that could not be loaded. The usual causes are missing `Upload` metadata, a file removed from storage, or a file owned by another work order. Re-upload the image from the report and generate the PDF again. The operations log shows `report_pdf_images_missing` with a count, without file names or personal data.

### Status and locking

A submitted report is locked. **Unlock for editing** requires a **Reason for unlocking** and is an administrator action.

### Customer portal and equipment history

Only submitted, locked reports can be opened. **View report** opens the PDF in the app, and **Download PDF** creates an attachment. An unlocked report is unavailable until it is locked again. The customer can **Acknowledge receipt** and **Request a contact information update**; the latter does not update the customer card automatically.

The report's units are linked to the customer's equipment. **Service history** on the asset shows every check with date, result, comments, checkpoints and images.

## Documents

The document library is at `/dokumenter` and is internal documentation with folders, tags, versions and full-text search.

### Folders and upload

1. Press **New folder**, or **Import folder structure** for a whole folder of PDFs.
2. Press **Upload PDF**, choose the **PDF file**, and enter **Title**, optional **Description** and **Tags**.
3. Press **Save document**.

Only PDFs are allowed. **Move to folder** and **Add tag** work on several documents at once. **Delete selected** is an administrator action.

### Tags, search and OCR

Document tags are separate from work order tags. Search **title, description, file name or tags**, and choose **This folder** or **Whole library**; **Filter by tag:** narrows further. Text is extracted on upload, and scanned PDFs are run through OCR. **Index all PDFs for search** fills in missing content.

### Versions and sharing

**Edit → Replace file (optional)** creates a new version. Previous versions are under **Previous versions ({count})** with **Download** and **Restore**.

The library is internal and controlled by **View documents** and **Manage documents**. To share with a customer, add the file to the customer card under **More → Attachments**; it appears in the customer's **Documents**. Work order attachments are internal. Reports are shown only when submitted and locked.

## Map, search, notifications and messages

**Map** ( `/kart` ) shows active work orders grouped by location. Filter by **Technician**, **From date** and **To date**. **Open in Maps** gives directions. Markers are placed at the address when geocoded; otherwise the work order is shown at the customer's city.

**Search** ( `/sok` ) finds customers, work orders, equipment, contacts, reports, comments, employees, checklists, folders and documents. Use the top bar, **Ctrl+K** (⌘K on macOS) or ***/***. Enter at least two characters. **Best matches** appears first, **See all ({count})** opens the full list, and **Filters** narrows by type, status, customer, technician, date, tag and HMS type. Search is fuzzy and tolerates typos; document hits open the document in the library. **Save this search** stores a saved search.

**Notifications** ( `/varsler` ) shows **{unread} unread of {total} total**, **Unread only** and **Mark all as read**. The types are **Assignment**, **Date change**, **Cancellation**, **Report**, **Booking**, **Service agreement**, **Overdue** and **Contact update**. **Settings** lets you choose **In app** and **Email** per event, and turn **Notification sound** on or off.

**Messages** ( `/meldinger` ) collects SMS conversations. Filter on **All** or **Unread only**, search by customer, contact or work order, and **Start conversation** from a work order. **Quick messages** sends **Send booking confirmation**, **Send reminder**, **Send on my way**, **Send completed** or **Request review**. **Consent and opt-out** shows **Consent registered** or **Not registered**; without consent no SMS is sent. Each outgoing message shows **Status** (Queued, Sent, Delivered or Failed). Threads are scoped to your department; portal users see only their own.

An administrator can configure **Automatic reminders** under **Settings → SMS**. Rules send SMS, email or both a chosen number of hours before the visit, with quiet hours and a maximum per visit. **Recent reminders** shows what is planned, sent or skipped — without the message text.

## Customer portal from the office

The customer sees **My Company: {customerName}** and the tabs **Overview**, **Bookings**, **Reports**, **Documents**, **Equipment**, **Contracts**, **Visits**, **Messages** and **Contact**.

1. Open the customer card → **More → Portal users**.
2. Press **Create and send invitation**, and enter name, email and contact.

3. Send the invitation; the user sets a password.
4. Press **Activate** once the password is set.

**Deactivate** revokes access immediately. A portal user must be linked to a contact at the same customer.

**Bookings** shows requests and confirmed appointments. **Reports** lets the customer acknowledge. **Documents** shows files you have shared through the customer card's **Attachments**. **Equipment** shows status, warranty, latest reading and plans. **Contracts** shows service agreements and coverage lines. **Visits** shows tracking links and reviews. The tracking page shows the planned time when the technician has not shared a location, and a live ETA only when the latest position is fresh (under ten minutes). If the position is older, it says explicitly how old it is; the page never pretends an old ping is live. The ETA refreshes automatically every 30 seconds. **Messages** lets the customer reply by SMS with consent.

The customer can **Book an available time**, **Request a new time**, **Cancel request**, **Acknowledge receipt** and **Request a contact information update**. Booking and cancellation notify the office; contact updates must be made manually on the customer card.

## Daily and periodic routines

**Morning:** Check **Overdue**, **Today**, **Unassigned**. Assign anything that is without a technician today, and check double bookings and visits without a technician.

**During the day:** Follow **In progress** and **Overdue**. Move visits in the calendar, use dispatch for sickness or urgent work, and answer customers in **Messages**. Register consent before sending SMS.

**End of day:** Check that today's work orders are **Completed** or have a follow-up. Review **Unsigned reports**, check tomorrow's visits, and clear **Bookings** and **Notifications**.

**Weekly:** Review **Overdue** and **Unassigned**, next week's **Capacity**, **Expires within 60 days**, **Reports** with **Deviations only**, and customer portal users.

**Monthly:** Review expired agreements and agreement lines. Deactivate portal users who have left, and review equipment without readings or with expired warranty. SLA policies, business hours, holidays and price lists are managed by an administrator; ask for changes when needed.

## Customer transfer (administrator only)

The **Customer transfer** tab is under **Settings** and is shown only to an active administrator who also has **Manage settings**. Office staff, technicians and portal users get `403` even if they call the route directly. The tab moves one or more customers between installations; it is not a backup.

### Export

1. Open **Settings** → **Customer transfer**, search for the customer and tick it.
2. Choose whether files, revision snapshots, deleted rows and legal hold should be included. Signature images are off by default and require a documented assessment.
3. Read the personal-data notice and tick the confirmation.
4. Select **Start export**. You get a ZIP with `manifest.json`, `checksums.sha256`, data files and files. The package is available for 24 hours and can only be downloaded by you.

5. Optionally verify manually with `sha256sum -c checksums.sha256`. The checksums detect corruption but do not prove who created the package.

The package never contains passwords, password hashes, 2FA secrets, sessions or API keys. Portal users follow as metadata and are imported deactivated without a password.

**Import**

1. Upload the ZIP. The server validates structure, checksums, file types and references, and shows a preview. Nothing is written yet.
2. Review counts, warnings and errors. Errors block the import. For conflicts, choose **skip** (default), **create new** or — for customer and equipment — **update**. Reports are never updated.
3. Tick the confirmation and select **Import the package**. All rows and files are written in one transaction; if anything fails, the database is unchanged and temporary files are cleaned up.
4. After import, each portal user has a new id, is deactivated and has no password. Send a normal invitation or reset the password before sign-in. Employees and templates are never created automatically.

**Repeated attempts and errors:** export is idempotent. The same administrator, the same customer selection and the same choices within ten minutes reuse the same job instead of starting a new one, so a double click does not consume another attempt. There are at most 10 export starts per hour; exceeding the limit gives a clear message to wait. Errors are stored on the job and shown in the interface, so you can correct the cause and try again without guessing. The preview is valid for 30 minutes; if the package or selection changed since it was shown, upload and preview again. Import errors block writing, and an aborted import rolls back the whole transaction and cleans up temporary files.

**Limits:** the package is limited to 500 MiB and 50 MiB per file, and the export runs synchronously. Global documents without a customer or unit owner are not included. Signatures, SLA clocks, bookings, notifications, SMS, tracking and report drafts are deliberately excluded. See `docs/backlog-customer-export-import.md` for the full list.

## Interface with the technician and administrator parts

| The office does | The technician does | Where it is described |
|---|---|---|
| Creates work orders with customer, contact, equipment and checklists | Takes the work order, opens the form and completes checkpoints | Technician part (Part 2) |
| Plans a visit with date, time and technician | Starts driving, marks **On site**, completes | Technician part |
| Reads reports, deviations and signatures | Signs, attaches images, submits the report | Technician part |

The administrator part describes **Employees & Roles**, **Roles & permissions**, **Departments**, **Checklists / Templates**, **SLA and business hours**, branding, API keys, audit log, legal hold and restoration. See the administrator part (Part 3).

# Common mistakes and troubleshooting

| Symptom | Common cause | Do this |
|---|---|---|
| Cannot find the work order | Wrong department, view or filter | Reset search/filter, check department |
| Cannot assign | Missing **Assign technicians**, locked or legal hold | Request permission, or reopen as administrator |
| Drag and drop did not change the date | Drag and drop assigns a technician | Move the visit in the calendar |
| Calendar is empty | No visits, filter or outside ±6 months | Reset filter, choose **Today**, check visit |
| Overlapping visit | Technician has several jobs at the same time | Check capacity and move one of the visits if needed |
| Report is locked | Submitted report | Ask an administrator to **Unlock for editing** |
| Customer cannot see the report | Not locked, or missing portal/document access | Lock the report, check portal user |
| Customer cannot book | Missing contact, active user or available time | Check contact, user, 24 hours and available time slots |
| SMS is not sent | Missing consent, phone, quota or dry run | Register consent, check number/settings |
| Agreement creates no work orders | Paused/expired, future start or job not running | Activate, check start date and job |
| SLA is not shown | No active policy | Link a policy to the agreement or department |
| Price list does not invoice | Reference data | Invoice outside the platform |
| Attachment is not in the portal | Work order attachments are internal | Add the file to the customer card → **More** → **Attachments** |
| Unit cannot be moved | Active contract lines, number conflict, cross-customer reference or invalid target | Read the blocker in the preview, clear the contract line, or choose the right customer/site |
| Upload is rejected | File too large, user/customer quota reached, or proxy limit | Check file size, `UPLOAD_QUOTA_*`, disk and the proxy body limit |
| Customer transfer does not start | 10 starts per hour used up, an existing job is reused, or the preview expired | Wait, follow the existing job, or upload and preview the package again |

PART 2

# Technician

The field app for the **Technician** role: sign-in, My work orders, work order details, checklists, units, offline drafts and submission. Version 0.8.9 (2026-11-04).

The bottom bar has **Today** ( `/mine-oppdrag` ), **Work orders**, **Notifications** and **More**. **More** opens the same menu as the hamburger button at the top. Language, notification sound, reports, customers/equipment and settings are in that menu.

## First time: sign-in, 2FA and language

### Sign in

1. Open `/logg-inn` .
2. Enter **Email** and **Password**.
3. Select **Log in**.

You land on **Today**. A wrong password gives “Wrong email or password.” An account that has not been activated gives “Your account has not been activated yet. An administrator must activate it before you can log in.”

The session normally lasts 12 hours. Local drafts remain when the session expires, but are not sent until you sign in again.

### Two-factor authentication

2FA is optional but recommended. If enabled, you go to `/logg-inn/2fa` after your password.

1. Open your authenticator app.
2. Enter the 6-digit **One-time code** for the company.
3. Select **Verify**.

If you get “Incorrect one-time code. Try again.”, wait for the next code. If you have lost your phone, select **Sign in with a recovery code**, enter one of the codes you saved when you enabled 2FA, and select **Sign in with code**. An administrator can reset 2FA.

You cannot change your password in the field app. Sign out and use **Forgot password** on the sign-in page, or ask an administrator to send a new link. 2FA and recovery codes are managed under **Settings** → **Security**.

### Change language

1. Open **Menu** with the hamburger button at the top, or select **More** in the bottom bar.
2. Scroll to the bottom. **Change language** is there.
3. Choose **Norsk**, **English**, **Polski** or **Deutsch**.

The choice is stored in the `NEXT_LOCALE` cookie and takes effect without changing the URL. Dates and numbers follow the language. The fixed labels in the report PDF remain Norwegian in 0.8.9.

## Install the field app (PWA)

The field app is installable. It opens in its own window, starts at `/mine-oppdrag` and is locked to portrait orientation.

- **Android/Chrome:** Browser menu → **Install app** or **Add to Home screen**.
- **iPhone/iPad/Safari:** Share icon → **Add to Home Screen**.
- **Chrome/Edge on desktop:** The install icon in the address bar, or the menu → **Install**.

There is no separate “Install” button inside the app. Installation uses the manifest and requires HTTPS. The service worker is registered in production builds only.

## Scan QR and barcode

1. Select **Scan** on **Today**, in the top bar or in the menu.
2. Select **Start camera** and allow camera access the first time.
3. Hold the QR code on the equipment label steady inside the frame. The app opens the unit or work order if you have access.
4. If the camera does not work, enter the code in **Code or reference** and select **Look up**. You can paste the full link from the label.

The camera requires the app to be opened over **HTTPS** (a secure address). On a plain `http://` address, such as a LAN address during development, the browser blocks the camera and the app shows a separate HTTPS message. Use the installed app or the `https://` address in production.

- **iPhone/iPad (Safari):** Allow camera access when Safari asks. If you declined, go to **Settings** → **Safari** → **Camera** (or **Settings** → **the app**) and choose **Ask** or **Allow**. Camera scanning works in iOS Safari even without the browser's built-in barcode API; the app then uses a fallback scanner.
- **Android (Chrome):** Select **Allow** on the camera prompt. If you declined, open the lock icon by the address → **Permissions** → **Camera**, or **Settings** → **Apps** → **Chrome** → **Permissions** → **Camera**.
- **Camera is busy:** Close other apps that use the camera and select **Start camera** again.
- **The code is not read:** Hold the label steadier, avoid backlight and make sure there is enough light. The app keeps looking and shows a separate message if the image cannot be decoded.

If the browser has no camera support at all, a message says so – but on iPhone and Android the fallback scanner should be available as long as the app runs over HTTPS and you have granted camera access.

The app accepts unit QR codes, unit IDs, work order references (for example `O-261103-ABCD1234`) and order numbers. An unknown code, missing access and repeated scans give an explanation instead of opening the wrong page. Access is checked again on the server; a scan never grants more access than you already have.

If the customer already has QR labels from another system, they work too: the office registers the existing value under **Existing QR codes** on the unit. You scan the label the same way as our own, and the app opens the unit if you have access. The value is an opaque string – the app never follows a link or fetches content from the internet.

### Before you lose the network

1. Open **Today** and every work order you will perform.
2. Open every report form you need, so the page enters the offline cache.
3. Wait until the draft status shows **draft saved**.
4. Take and upload photos while online; a required photo field is not complete until the photo is uploaded.
5. Open any report PDFs you need while online.
6. Do not clear browser or app data.

The page cache lasts eight hours; local drafts last seven days.

## My work orders / Today

`/mine-oppdrag` is the home screen. At the top you see your name, **On duty**, **Online** or **Offline**, today's date, the number of jobs today, the number of **Overdue** jobs, and an estimated route distance when today's work orders have coordinates. If coordinates are missing, it says **Calculating route** or nothing – the app does not invent a distance.

The search field searches title, description, order number, customer name, city, address and contact person. **Search** runs it; **Reset** clears it.

### Next job

The **Next job** card shows the first remaining job today, or the next upcoming job if today is empty. It shows date and time, or **Time not set** / **All day**, title, customer, order number, address, contact person, **Job 1 of N**, and **Start form**, **Continue form** or **Open job**, **Call** and **Map**.

**Start form** opens the first checklist without a locked report. **Continue form** is used when a draft or unlocked report exists. If the work order has no template, the button says **Open job**.

### Overdue, available and later

- **Overdue** shows your open work orders with a scheduled date before today, oldest first.
- **Available** shows open work orders without a technician. With update permission, each card gets **Take work order**; you are assigned and see “Work order claimed and now under your work orders.” Without permission the button is hidden. Completed or cancelled work orders cannot be claimed.
- **Later today** shows the rest of the day, **Today's route list** shows the full list, and **Completed work orders / History** shows completed jobs with **Show more**.

### Job cards and status chips

The card shows title, status chip, customer, order number, contact, address, tags, date and time. A draft shows **Draft 14:32 · Continue**. **Call** and **Map** start the phone or Google Maps without opening the work order.

| Chip | Meaning |
|---|---|
| **Open / New** | Created, not planned or started |

| Chip | Meaning |
|---|---|
| **Waiting to be planned** | The system's starting status; not selected manually |
| **Planned** | Has a planned date or visit |
| **In progress** | Work is underway |
| **Waiting for customer** | Work is waiting for the customer |
| **Completed** | Closed and locked |
| **Cancelled** | Closed; not shown in the field app |

### Notifications

The **Notifications** tab in the bottom bar opens `/varsler`, with **Unread only**, **Mark all as read**, **Mark as read** and **Open**. The bell at the top shows the unread count. In the menu you can turn notification sound on or off with **Turn on notification sound** / **Turn off notification sound**.

### No jobs?

If it says **No active work orders in your calendar right now**:

1. Select **Reset** if a search is active.
2. Select the refresh icon at the top.
3. Check **Available** – there may be work orders you can take.
4. Check that you are in the right department; department scoping hides other departments' work orders.
5. Contact the office if the job is missing a technician or has the wrong date.

## Work order details

Open a card to reach `/mine-oppdrag/<work-order-id>`. At the top is a dark job panel with status chip, title, customer, address, department, order number, planned date, assigned technicians, contact person and checklist count.

### Customer, contact, address and map

**Quick actions** give you **Call contact**, **Map**, **Customer**, **Equipment** and **Documents**. **Call contact** uses the work order contact first, then the customer contact or main phone. **Map** opens the device's map app: Apple Maps on iPhone/iPad, Google Maps or the chosen map app on Android, and Google Maps in the browser on a desktop. It uses coordinates when available, otherwise a URL-encoded address, and is hidden when both are missing.

**Copy address** puts the full address on the clipboard. **Location and contact** shows the full address, company phone, the work order contact and the customer's other contacts with phone and email. If data is missing, it says **Not provided** or **No specific contact person is registered on this work order or customer.**

### Visit: start driving, on site, complete

The visit card shows **Today's visit** or **Next visit**, time range, status and technicians. The flow is: **Start driving** → **En route**, **On site** → **On site**, **Complete** → **Done**. Each tap records the time on the server. Tap only when you actually start driving, arrive or finish. After **Done** the visit is locked and it says **The visit is finished.**

With live tracking enabled, you can share your position while **En route**: **Share position** / **Stop sharing**. The position is deleted after 24 hours.

### Status, work description and comments

**Work order status** lets you choose a new status and select **Update**. **Complete** requires confirmation: “Complete this work order? The work order and report are locked for editing.” Cancellation is under **More actions** for roles with edit permission. Completed work orders are locked; reopening is done by an administrator.

**Work description / Instructions** is shown as Markdown. **Comments and updates** is a Markdown log: **Write comment** → **Add comment**.

### Attachments, personnel and history

**Attachments** handles photos and PDFs via **Add files**. Files upload one at a time; if one fails, its name is shown and you can try again.

**Assigned personnel and office** show who is on the work order. **Remove me** removes you, with **Undo** in the toast. **Add a colleague or office for follow-up** is used when someone else needs to see or take over.

**Activity log** shows status changes, assignments, comments, reports and locking. Service history lives on the individual unit; see “Units at the top of each checklist”.

### Checklists and reports

**Checklists for the work order** lists every template, with HMS type, **Report no.** when one exists, **Locked**/**Unlocked**, **Fill in form** →, **Edit** when unlocked for correction, and **View report (PDF)** when a report exists.

The progress ring shows answered checkpoints, deviations and draft status. **Do you need an extra form?** adds a template you have access to; otherwise the office must do it.

If it says **No checklists are linked to this work order yet**, you cannot fill in anything until the office adds a template.

## The report form

The form opens at `/mine-oppdrag/<work-order-id>/skjema/<template-id>`.

The fixed step strip shows **Checkpoints**, **Photos** and **Sign**, plus draft status and **Save draft now**.

### Locked report and draft restore

If the report is submitted and locked, the page shows **The report is locked**. You can open **View the report (PDF)** or go back. Only an administrator can unlock it for correction.

If a saved draft exists, the form opens with the fields as you left them, and you see “📝 A saved draft has been restored with all fields as you left them.” A submitted report is never overwritten by a local draft.

## Units at the top of each checklist

Each unit is a block; details sit at the top:

- **Get from the customer equipment** – choose a registered unit; designation, serial number and location are filled in.
- **New unit (not registered before)** – when the equipment is not on the customer record.
- **Designation** * – required; enter what you see, for example “North gate”.
- **Serial number** and **Location** – optional, but important for history.

A linked unit shows 🔗 **Linked to a registered unit on the customer record**. A new unit with a name shows ＋ **Saved as a new unit for the customer**. On submission, the unit is created or reused on the customer.

**Each registered unit can only be used once in the same report.** A duplicate choice is rejected with “This unit is already selected in another unit row. Choose a different unit.”

**Add unit** adds a row; **Remove unit** confirms and deletes its checkpoints. Use **New unit** only for new equipment, not to bypass the duplicate check.

Units provide service history, the customer's equipment list, the office overview and the basis for QR labels.

## Checkpoints

Each checkpoint comes from the template:

- **Status choice** – OK / Deviation / Correction needed / Not applicable. A follow-up choice opens a required comment: “💬 Comment for “…” – required”.
- **Checkbox** – must be checked when required.
- **Free text / text area** – the text area supports Markdown.
- **Number, date and dropdown** – ordinary required fields.
- **Photo** – **Take photo** / **Choose photo**. Compressed and uploaded immediately; offline they stay **Ready to upload** or **Upload failed** and retry when online.
- **Signature** – a text field where you enter your full name.

Fields marked `*` need a real value; a checkbox must be checked and a photo field needs an uploaded photo.

**Conditional fields** appear only when another field has a specific value. If you change the source, the field may disappear. Answer every visible required field; hidden fields are not required.

Follow-up shows “One checkpoint requires follow-up” or “{count} checkpoints require follow-up”. If the unit is still **Approved / done**, the app asks you to confirm, and you can select **Set the result to “Deviation found”**.

Photos are evidence: photograph the deviation, meter, damage or completed correction.

### Signatures

**Signatures** at the bottom are optional – technician, customer or both. Draw with a finger or mouse, use **Full screen**, and **Clear** to start again. **Technician** is pre-filled with your name; **Customer** with the work order contact.

An earlier signature is kept if you do not sign again: **Kept if you do not sign again.** If you enter a name, it must be one of the work order's contact persons or technicians. Otherwise submission is rejected with “The signer name must be one of the work order's contact persons or technicians.”

### Markdown in notes and comments

Free-text fields, unit comments, general notes and work order comments use Markdown. The toolbar has bold, italic, strikethrough, highlight, heading, lists, task list, quote, callout, code, link, table, footnote and hidden comment. **Preview** shows the result.

Markdown is sanitised on the server and rendered the same way in the browser and the PDF. Use **General notes for the job** for the overall picture, and **Comment for this unit** for what applies to one unit.

### Mobile swipe-to-complete

On mobile you can approve a unit with a swipe:

1. Swipe **right** on the unit header. You see a percentage and **keep swiping**.
2. At the threshold it says **Release to complete**.
3. Release. The unit is set to **Approved / done**.

The swipe requires 84 pixels; vertical movement scrolls and cancels. The fallback **Complete unit** button is at least 48 pixels high and does the same.

After approval you get **Undo** for three seconds: “Unit N is approved. Undo is available for 3 seconds.” **Undo** restores exactly the previous status.

Swipe and button are blocked when the unit is **Approved**, set to **Deviation found**, missing **Designation**, missing a required answer, photo or checkbox, or has a status choice that requires follow-up (“Cannot complete: “…” is set to “…”. Use the Approved/Deviation options.”).

Therefore a quick gesture cannot approve an incomplete or deviating unit. The swipe uses the same validation as the form and the server.

## Submit and sign

### Review

**Sign and submit report** opens **Sign and submit the report?** with **Units: N**, **Deviations: N**, **Photos: N**, **Signatures: technician … , customer …**, and **When you submit, the report is locked and cannot be changed unless an administrator unlocks it.**

Clear **Mark the work order as completed** if the work order should stay open. It is checked by default and sets the work order to **Completed** on submission.

Select **Sign and submit report**, or **Back to the form** to correct something.

### Submission requirements

Submit when online, the template belongs to the work order, you have `reports.write` in the right department, and the report is not locked or under legal hold. When the template requires units, the report must have at least one unit, and every unit must have a **Designation** and be assessed as **Approved / done** or **Deviation found**.

Answer every visible required checkpoint; upload required photos and check required boxes. The same registered unit cannot appear in several rows. Cancelled work orders cannot be completed by technicians. Any signer names must be people linked to the work order.

A signature is optional; the report can be submitted without one. If an administrator has unlocked the report, it locks again automatically when you save.

### After submission

The server confirms with **The report has been submitted and locked**, **Open report (PDF)**, **Open work order** and, in the field app, **Next assignment**.

The report is locked, local draft and outbox entries are cleared, and the form page leaves the offline cache. The office sees the report, the customer can see the submitted report and acknowledge it, and the units update service history. If the report has deviations, one undated remediation work order is created with you as technician.

If **Mark the work order as completed** is checked, the work order is set to **Completed** in the same transaction. Correction requires an administrator to unlock the report with a reason. A new submission creates a revision and resets any customer acknowledgement.

### PDF in the app or as a download

**View report (PDF)** opens the report in the app's own PDF viewer. There you can scroll, zoom and close without leaving the work order. **Download PDF** creates a file you can send or save. A direct link to the PDF route always downloads, because only the app's own viewer asks for inline display.

The PDF is sanitised before it is shown. If it contains active content that cannot be removed safely, the viewer shows an error with **Try again** and download instead of opening anything unsafe. The report data and the lock are unchanged. If an image or signature is missing, the PDF shows a warning with the number of images that could not be loaded; re-upload the image from the report and generate the PDF again.

Open any PDFs you need while online. PDF viewing requires a connection, and a direct download should be made before you lose coverage.

## Offline, drafts and the outbox

### Autosave

The form saves locally about 200 ms after a change, on the server after about 1.2 seconds, when the tab is hidden or closed, and on **Save draft now**.

The local draft contains units, answers, conditions, notes, names and signatures, and survives network loss, tab closure and restart. The server draft follows the user to another device. If the local draft is newer, it is restored when the form opens.

Drafts are per user, work order and template, and expire after seven days.

### The offline banner

The banner at the top is visible when you are offline or have pending drafts:

- 📴 **You are offline – drafts are stored on this device and sent when the app is open and the network returns**
- 📴 **Offline · {count} draft(s) waiting**
- 📤 **{count} draft(s) waiting to sync**
- ⚠️ **{count} draft(s) could not be sent**
- 🔄 **Syncing drafts …**
- ✓ **{count} sent**

**Open the draft list** goes to `/offline.html` with stored pages, drafts, failed sends and **Discard my drafts**.

### Outbox and background sync

Only the newest draft per report is kept. The outbox is the browser's local send queue, the payload is the content sent to the server, and backoff is the waiting time between retries.

- **Chromium** can use Background Sync: the browser wakes the service worker and sends even without an open tab.
- **Safari/Firefox** do not support Background Sync. Drafts are sent when the app is open and the network returns, when the tab becomes visible, or by the 30-second timer. Keep the app open until ✓ **sent**.

Backoff runs 15 seconds to 30 minutes, maximum 10 attempts, then terminal **could not be sent**. Payloads over 2 MB are rejected; 400/413 are terminal; 429/5xx stop the queue and back off.

### 401/403 and expired sessions

A 401/403 means the session or access is gone, not that the draft is broken. The service worker stops replay, page caches and the offline marker are cleared, and the draft/outbox entry is kept. Sign in again, then select **Send now** or save the form. A 403 can also mean lost permission; the office must restore access.

**Discard my drafts** deletes all local drafts and pending uploads for the user on the device. Use it only when sure, and always on a shared device before leaving.

## Safety and quality in the field

- **Swipe or “Complete unit” is only for fully documented, non-deviating units.** Use **Deviation found** when something is wrong, missing or needs correction.
- **Never approve something you intend to fix later.** “Approved” means checked and in order now; correction belongs in the deviation or a separate work order.
- **Describe the deviation concretely:** what, where, measurement and what must be done. Attach a photo when relevant.
- **Use the recorded timestamps.** Start driving, on site and done are recorded when you tap; do not tidy them up afterwards.

- **Check unit identity.** Correct designation, serial number and location make the history useful; one registered unit is used once per report.
- **Only sign what you have reviewed.** A signature is optional, but the name must belong to a person on the work order.
- **The report locks on submission.** Corrections require an administrator unlock and are logged as a revision.
- **Legal hold stops everything.** A frozen report cannot be submitted or changed.
- **Do not share your sign-in.** Drafts and the page cache are per user; sign out and discard drafts on a shared device.
- **Do not invent time, distance or data. Time not set** means the time is not set.

## Common field problems

| Problem | Likely cause | Fix |
|---|---|---|
| **No active work orders** | Search, wrong department, no assignment, or no jobs | Reset search, refresh, check **Available**, ask the office |
| **The report is locked** | The report is submitted and locked | Open the PDF or ask an administrator to unlock it |
| **No checklists** | The template has not been added | Ask the office to add it. If you have access, use **Do you need an extra form?** |
| **Upload failed** | Network, large file, camera/storage | Check network, **Try again**, or use **Choose photo** |
| **Draft waiting / sync stuck** | Offline, expired session, 429/5xx or terminal | Get online, keep the app open, **Send now**. On 401/403, sign in. On **could not be sent**, **Try again** or discard |
| **Wrong or duplicate unit** | Wrong row or duplicate registered unit | **Remove unit**, add the right one. For duplicates, choose another or **New unit** |
| **Cannot submit** | Missing unit, assessment, required answer, photo, checkbox or name; locked; offline; no checklist | Read the message, correct the field, try again. **Missing assessment** means **Approved** or **Deviation** per unit |
| **Cannot complete with swipe** | No name, deviation, or missing required field | Fill in the field, upload the photo, or use **Deviation found** |
| **2FA code fails** | Wrong code, clock drift, used code | Wait for a new code, check the clock, use an unused recovery code, or ask for a reset |
| **Visit buttons missing / status locked** | Visit or work order closed, or no permission | Contact the office; only open visits and work orders can be changed |
| **PDF or offline page missing** | No network, report, access, cache, or active content that cannot be sanitised | Get online, open **View report (PDF)**, download the file, reopen the job online |

| Problem | Likely cause | Fix |
|---|---|---|
| **Image missing from the PDF** | File deleted, missing metadata, or owned by another work order | Re-upload the image from the report and generate the PDF again |

## How technician work connects to the rest

| Area | What you do in the field | Where others see it |
|---|---|---|
| **Office** | Update status, comment, submit reports | `/oppdrag`, `/rapporter`, dashboard, notifications. The office assigns, reschedules and adds templates; an administrator unlocks reports. |
| **Customer portal** | Sign and submit a report | The customer sees submitted, locked reports in `/kundeportal/rapporter`; PDF download and **Acknowledge receipt** |
| **Equipment module** | Choose or create a unit, record serial and location | Service history, timeline, readings, QR label on `/enheter/<id>` and **Equipment** tab |
| **Administrator** | Follow template requirements, use right department and user | Manages **Checklists / Templates**, **Employees**, departments, permissions, 2FA and unlocks in **Settings** |

PART 3

# Administrator

Applies to FSM Platform **0.8.9** (2026-11-04). This chapter is for system administrators who configure the platform, manage users and permissions, set up integrations, and are responsible for data, security and operations. It covers configuration and governance. Day-to-day workflows for office, technician and customer portal users are described in their own chapters.

## 1. The administrator role and delegation

Administrators always have every permission, including deletion, unlocking submitted reports, legal hold, roles, departments and API keys. The Office role is an operations role with broad access inside its own department, but without the most security-sensitive actions.

| Task | Administrator | Office | Recommendation |
|---|---|---|---|
| Roles, departments and API keys | Yes | No | Keep with administrators. |
| Administrators, deletion and moves | Yes | No | Keep with administrators. |
| 2FA, report unlocking and legal hold | Yes | No | Keep with administrators. |
| Activity log | Yes | No | Grant `audit.view` deliberately. |
| Employees in own department | Yes, all | Yes, own department | Delegate to the department lead. |
| Customers, work orders, agreements, templates and reports | Yes | Yes | Delegate to office and subject-matter owners. |

Office has `settings.manage`, `users.manage` and `dispatch.configure` by default, so it can change branding, map, SMS and dispatch settings. If that is too broad, create a separate role, for example **Service manager**, without `settings.manage`. Changes to the built-in Office role affect everyone who has it.

**Least privilege:** grant the weakest role that solves the task, use dedicated roles for leads and integrations, and keep at least two active administrators.

## 2. Departments and department scoping

Work orders, service agreements and employees belong to a department. Employees only see work orders in their own department. Customers and documents are shared; reports are department-scoped unless the user has `reports.viewAllDepartments`.

**Create a department:**

1. Go to **Settings** → **Departments**.
2. Enter **Name** and optional contact details.

3. Leave **Enable live tracking** off until privacy has been assessed.
4. Select **Create department**.

**Expected result:** the department appears with counts for employees, work orders and agreements. One department is the **Default** and is used when a work order or booking does not specify another. The default department cannot be deactivated until another department is set as default.

**Warning:** deactivating a department signs out all employees in it, clears calendar subscriptions and makes its work orders unavailable, because work-order queries require an active department. Accounts are kept and the department can be reactivated. Notify users in advance.

# 3. Users, invitations and sessions

**Create a user:** go to **Settings** → **Employees & Roles** or `/brukere`, select **Add new user**, enter **Full name**, **Email address** and optional **Phone**, choose **Role / Access level** and **Department**, then select **Create and generate invitation**. Employees need a department to see and create work orders.

The user is created inactive. You either get confirmation that the invitation email was sent, or an **Invitation link** you can send manually. The link is shown only in the page content, is valid for seven days, and can only be used by an inactive account. Without `SMTP_HOST` and `SMTP_USER`, the app cannot send email. Active users must use **Forgot password**.

**Activate the account:** the user opens the link and chooses a password. The account is still inactive. Find the user under **Employees & Roles** and select **Activate**. Only then can the user sign in. Ask the user to set up 2FA under **Settings** → **Security**.

**Deactivate and delete: Deactivate** signs the user out, clears the calendar subscription and blocks sign-in and API access. **Activate** restores access. **Delete user** is administrator-only and cannot be used on yourself or the last active administrator.

**Passwords and 2FA:** administrators cannot see or set a password. **Forgot password** sends a one-time link valid for one hour, rate-limited per email and IP. **Reset 2FA** disables two-factor authentication, deletes recovery codes and signs out all sessions. Use it only when a device is lost or misuse is suspected.

**Sessions:** there is no session list or “sign out everywhere” button. Sessions are invalidated when the password changes, 2FA changes or is reset, the user is deactivated, or the user’s role, permissions or department change. Editing a role definition applies on the next request without signing users out.

## Change password, 2FA and recovery codes

1. To change your password, sign out and use **Forgot password** on the sign-in page. 0.8.9 has no separate change-password page. The link is valid for one hour, and all other sessions are invalidated when the new password is set.
2. Go to **Settings** → **Security** to turn on **Two-factor authentication**. Scan the QR code with your authenticator app and enter the 6-digit code.
3. Store the **Recovery codes** in a safe place. They are shown only once, and each code can be used once. **Create new codes** invalidates the old ones.
4. Use **Log out** on shared devices. If misuse is suspected, change the password via **Forgot password** and reset 2FA.

An administrator can reset a user’s 2FA but cannot see the password. A reset deletes the recovery codes and signs out all sessions.

### SSO and SCIM

SSO is configured with `SSO_PROVIDERS` and the `SSO_<SLUG>_*` variables, not in the interface. Entra ID, Keycloak, Authentik and Zitadel are supported with OIDC, PKCE, state/nonce and verified email. The redirect URL is `https://<NEXTAUTH_URL>/api/auth/callback/sso-<slug>`. After a restart, **Settings → SSO and SCIM** shows the provider status, issuer, default role and allowed domains. Password sign-in always remains as break-glass.

SCIM 2.0 is served at `/api/scim/v2` with a dedicated Bearer token created, rotated and revoked in the same tab. `DELETE` deactivates the user without deleting it, and `ADMIN` requires both an explicit group mapping and `SCIM_ALLOW_ADMIN=true`. Use one token per integration, and keep at least two local administrators with strong passwords and 2FA.

## 4. Roles and permissions

| Role | Key | Default content |
|---|---|---|
| **Administrator** | `ADMIN` | All permissions; the list cannot be changed. |
| **Office** | `OFFICE` | Work orders, customers, agreements, equipment, reports, documents, templates, employees and settings in its own department. |
| **Technician** | `TECHNICIAN` | View and create work orders, update status, view customers, agreements and equipment, fill in reports, view documents and templates. |
| **User** | `USER` | No permissions by default. |
| **Customer** | `CUSTOMER` | The customer portal, controlled by the customer link. |

Administrator and Customer are locked. Office and Technician can be edited, but system roles cannot be deleted.

**Custom role:** go to **Settings → Roles & permissions → New role**, enter **Name**, optional **Description**, select **Permissions** and choose **Create role**. The role gets a fixed key based on its name; the key does not change if the display name changes. A role in use cannot be deleted until its users are moved.

**Per user:** open **Permissions for this user**, select the list and choose **Save permissions**. The list overrides the role completely; if it matches the role’s list, the override is removed. **Use the role's permissions** does the same. Customisation does not apply to administrators or portal users, and changing a role clears the list.

The permission groups are **Work orders** (`workorders.*`, `dispatch.configure`), **Customers and equipment** (`customers.*`, `equipment.*`), **Agreements and SLA** (`contracts.*`, `sla.manage`), **Reports and documents** (`reports.*`, `documents.*`) and **Administration** (`templates.*`, `users.*`, `settings.manage`, `audit.view`).

**Test a role:** use a dedicated account, try a direct URL it should not have (for example `/innstillinger?tab=roller`) and one mutation, confirm both are rejected, then deactivate. Authorisation is enforced on the server; hidden buttons are not security.

**Common mistakes:** giving a broad role to everyone, forgetting a per-user override, deleting a system role or a role in use, and giving Office `settings.manage` when it is not intended.

## 5. Customers and customer portal users

Customers are created under **Customers** → **New customer**. The customer card has **Overview**, **Work orders**, **Agreements**, **Equipment** and **More**. Portal users are created under **More** → **Portal users**.

**Create a portal user:**

1. Create the customer and at least one **Contact** with an email address.
2. Go to **More** → **Portal users**.
3. Enter **Name**, **Email** and choose **Contact**.
4. Select **Create and send invitation**.
5. Send the link manually if the email fails.
6. Select **Activate** after the user has chosen a password.

The user receives the `CUSTOMER` role, a customer link and a contact link, and remains inactive until you activate the account. **New link** only works for inactive accounts.

The portal shows the customer's own overview, bookings, reports, documents, equipment, agreements, visits, messages and contact details. Reports appear only when submitted and locked; documents only when shared with the customer (`CUSTOMER`-owned).

Portal users never see other customers, internal notes, internal permissions, drafts or unlocked reports. `/uploads/...` is checked per file, and PDF responses are `no-store`.

**Security:** use separate accounts per person. **Deactivate** when someone leaves; this signs them out, clears the calendar subscription and revokes portal access. Deleting the customer deactivates all portal users, invalidates sessions and closes booking. After restoring, each portal user must be activated again. A portal user must always have a valid contact on the same customer.

**Delegation:** the portal has server-enforced capabilities for booking, documents, agreements and equipment, but 0.8.9 has no form for editing them. A contact without a delegation row gets full portal access; restricted access must be configured and tested with the actual account.

## 6. Checklists and templates

**Create a template:**

1. Go to **Settings** → **Checklists / Templates**.
2. Select **+ Create**, enter **Checklist name** and choose **Category**.
3. Select **Edit**, add fields under **Add a new checkpoint**, and set **Required field** where needed.
4. Use **Move up**/**Move down**, then select **Save template with logic**.

Field types are **Short text**, **Long text** (Markdown), **Checkbox**, **Dropdown**, **Condition + comment** (OK/Deviation/To be fixed/Not applicable), **Number**, **Signature**, **Image upload** and **Date**.

**Required and conditional:** required fields must be answered before submission. **Condition (If This Then That)** shows a field only when another field has a specific value, for example `Deviation`. **A comment is required for these options** applies to condition fields; the default is `Deviation` and `To be fixed`. Clear it to never require one.

**Units and report name: Link the report to equipment units** is on by default and gives one assessment per unit. Turn it off for reports that concern the whole work order. **Report name** appears in the PDF and report lists; an empty field uses the template name.

**Category and HMS: Service** or **HMS form**. HMS requires an **HMS type**: general, SJA, hot work, internal control, deviation, safety round or written instruction. Category and HMS type are frozen at submission.

**Linking and versioning:** templates can be linked to work orders, service agreements and maintenance plans. At submission, a snapshot of the template, report name, unit requirement, category and HMS type is stored. Later changes do not affect existing reports, even when unlocked; new fields apply only to new reports.

**Delete a template:** administrator only. A template cannot be deleted while it is used by reports, work orders or active agreements. A blocked attempt is logged.

### HMS library and QR labels

The **HMS library** is at `/hms`. It groups HMS forms by type and lists templates and completed HMS documentation. **Link HMS form to work order** chooses a form and an active work order; the form is added to the work order and completed there. HMS templates are managed under **Settings → Checklists / Templates** with category **HMS form**.

**QR label:** open the unit at `/enheter/<id>`, go to **QR label** and use **Rotate QR token** if the label is lost or shared. Print the label from `/enheter/<id>/etikett`. A rotated token invalidates the old label.

## 7. Work orders and reports

The statuses are **Open / New** (`OPEN`), **Waiting for planning** (`PENDING`, the system start status), **Planned** (`PLANNED`), **Waiting for customer** (`WAITING`), **In progress** (`IN_PROGRESS`), **Completed** (`COMPLETED`) and **Cancelled** (`CANCELLED`).

Office and technicians with `workorders.update` can change open statuses freely. Completed and cancelled are closed; only an administrator can reopen them, with a reason. Cancellation requires `workorders.edit`. The report has its own lock and is not affected by the work order status.

**Unlock a report:** open the work order, find the checklist, select **Unlock for correction**, enter **Reason for unlocking** and confirm. The report becomes **Unlocked**, can be edited, and a revision of the previous version is stored. The customer’s acknowledgement is withdrawn. On resubmission, the report is locked automatically and gets a new revision. Legal hold blocks unlocking.

**Revisions:** every unlock and submission creates a revision that freezes the form, units, comments, signatures, report number, category, HMS type and acknowledgement. **Revisions** shows the content and lets you download it as a PDF.

**Report numbers:** HMS reports receive an automatic `abc1234` number (three lowercase letters and four digits) on first submission. It is kept across revisions. Ordinary service reports have no automatic number.

**PDF and branding:** the PDF uses the company name, logo and primary colour from **Branding**, is stored as a `WORK_ORDER` upload and requires `reports.view`. The response is `no-store`. Branding is part of the PDF hash, so changes apply next time the PDF is generated. Report PDF labels are Norwegian in 0.8.9; agreement PDFs support all four languages. The generated remediation work order title is also Norwegian (“Utbedring av avvik – …”) in 0.8.9.

**Legal hold and retention:** legal hold prevents hard deletion and freezes the report. On a work order, all reports inherit the flag. Releasing it requires a reason and does not release the reports’ own flags; **Report hold** must be released per report with its own reason. Soft-deleted work orders can be hard-deleted after `RETENTION_DAYS` (default 365) only when they have no reports. **Recycle bin** and **Restore** are administrator-only.

## 8. Service agreements, SLA and price lists

**SLA policy:** go to **Settings → SLA and business hours**, enter **Name**, **Department** or **All departments**, **Response time (minutes)**, **Resolution time (minutes)** and **Count business hours only**, then select **Create policy**. The work order’s agreement can point to a policy; otherwise an active customer agreement is used, then the department default and finally the global policy. The clock starts at creation, response is recorded on `EN_ROUTE` / `ON_SITE`, resolution on `DONE` / `COMPLETED`, and `CANCELLED` cancels it. States: `ON_TRACK`, `AT_RISK`, `BREACHED`, `MET`, `CANCELLED`.

**Business hours and holidays:** choose **Department**, select **Show**, set **Opens**/**Closes** or **Closed** per weekday, and select **Save business hours**. Add **Holidays** with date, name and optional department. Global holidays apply to all; department-specific rows override only that department. Without business hours, weekdays 08–16 in Europe/Oslo are used. All deadlines are calculated through this calendar.

**Service agreement:** create it from the customer card, **Agreements** tab, or via **Agreements**. Enter **Title**, **Start date**, **Interval** (3, 6, 12 or 24 months), optional **Contact**, **Checklist** and **Department**, then select **Create service agreement**. The agreement becomes **Active** (or **Draft** if selected) with a next service date. A start date today creates the first work order immediately; a past date uses the next future interval. Each interval creates a work order with customer data, a checklist and the previous report as a PDF.

**Coverage:** an agreement without lines covers all equipment. A line without a product also covers all; a line with a product covers only that product. A line can have **Coverage**, **Visits/year**, **Response (min)** and **Resolution (min)**.

**Renewal: Automatic** renews after the end date and generates a new work order. **Manual** becomes **Expired** after the end date. **Notice (days)** controls the expiry notice. Automatic renewal runs exactly once even with multiple instances.

**Price lists:** create them under **SLA and business hours → Price lists** with **Code**, **Description**, **Unit** and **Amount**. They are reference data for agreement PDFs and settings; the system never calculates, invoices or exports amounts.

## 9. Booking, SMS, tracking and reviews

**Booking:** the customer chooses a requested date (up to 60 days ahead) and writes the desired time in the description; the customer cannot select a time block. The request creates a new work order with status **Open/New** and no visit or confirmed time. The office or technician plans it and sets **Planned**, or **Waiting** if

the time does not fit. **Bookings** shows unconfirmed requests; they can be declined, but planning happens on the work order.

**SMS:** the provider is selected with `SMS_PROVIDER`, not in the UI. The default `console` is a dry run; `linkmobility`, `pswincom` and `twilio` are supported. **Sender** comes from `SMS_SENDER_ID`. Under **Settings → SMS and messaging** you can see the provider, sender, dry-run state and **Webhook signature**, and set **Default language**, **Opt-out keywords**, **Opt-in keywords**, **Monthly cap** and **Templates per event**.

SMS is sent only when the recipient has `optedIn = true` and `optedOut = false`. STOP/START is enforced from the keywords, and consent is checked when queuing and before sending. `SMS_ALLOWLIST` can restrict test and staging. The webhook at `/api/webhooks/sms/<provider>` requires an HMAC or Twilio signature; without `SMS_WEBHOOK_SECRET`, deliveries are rejected (except console in development). Outgoing SMS is a queue with three attempts; provider delivery status updates **Sent**, **Delivered** or **Failed** in the same conversation without creating a new message. Conversations appear under **Messages**.

**Reminders:** under **Settings → SMS → Automatic reminders** you create rules per department. A rule has a **Channel** (SMS, email or both), **Hours before visit** (for example 24 or 2), **SMS template**, **Quiet hours**, **Time zone** and **Maximum reminders per visit**. The worker sweeps every five minutes, skips customers without consent, respects quiet hours, and cancels the reminder if the visit is moved, cancelled or completed. **Recent reminders** shows planned, sent and skipped rows with a reason — without message text or phone numbers. Sending happens in the background, so a slow SMTP or SMS provider never blocks a page.

**Tracking:** off by default per department. Enable it under **Departments → Edit department → Enable live tracking** after a DPIA, because the technician's location is personal data. An assigned technician can share location while the visit is `EN_ROUTE`. The public link shows only first name, status, ETA, planned time and last updated — never coordinates, a map or history. Pings are kept for 24 hours and deleted on `DONE` / `CANCELLED`; the link expires two hours after the visit, and there is one active link per visit. `TRACKING_AVERAGE_SPEED_KMH` (default 50) controls the fallback estimate. With `OSRM_BASE_URL`, an external routing provider is used for ETA; without it, ETA is calculated locally. A ping older than ten minutes is shown as **last known position**, not as live. The public ETA endpoint is polled every 30 seconds and has its own per-token rate limit.

**Reviews:** internal and never published. The customer can request a review in the portal after a completed visit, or the office can send an SMS with "Request review". A review requires a completed visit, can be submitted once and is stored with a hashed token. Reviews are not shown as a separate dashboard in 0.8.9.

## 10. Branding, map, language and notifications

**Dispatch and workforce:** under **Settings → Dispatch**, configure **Assignment rules** (weights for skills, distance, travel time, availability, SLA and load) and **Territories**. Weights are relative. **Capacity plans** are shown under **Calendar → Capacity** but are not edited in **Settings** in 0.8.9. Skills, absence and working hours feed the suggestions but also have no separate form in **Settings**; keep the data updated outside the interface.

**Branding:** under **Settings → Branding**, set **Company name**, **Primary colour (hex code)** and **Company logo (optional)**. The logo must be uploaded through the app; external URLs are rejected. It is used in reports, email and the dashboard header, and applies to office and customer portal, not the dark field app. Customers can have their own colour and logo in the portal.

**Map and geocoding:** under **Settings** → **Map**, choose a **Map provider**. **OpenStreetMap (no key)** is the default; **MapTiler** and **Mapbox** require an **API key**. The key is stored on the server and sent only to the map in the browser; use a domain-restricted key. Set **Default layer**, **Default zoom (2–19)**, **Center** and **Enable scroll wheel zoom**. Geocoding is on by default (`GEOCODE_ENABLED=true`) via OpenStreetMap/Nominatim; set `GEOCODE_USER_AGENT`, or disable with `GEOCODE_ENABLED=false`.

**Language:** Norwegian, English, Polish and German have complete catalogues. The top-bar selector stores the `NEXT_LOCALE` cookie and uses `router.refresh()`, without a URL prefix. Dates and numbers follow `nb-NO`, `en-GB`, `pl-PL` and `de-DE`. SMS uses the recipient contact's `locale`; email notifications use the user's `locale`. Reports have Norwegian PDF labels in 0.8.9; translations live in `messages/` and are checked with `npm run check:i18n`.

**Notifications:** in-app and email. Each user chooses events and channels under **Notifications** → **Settings**; there is no global default. Make sure SMTP is configured. The `Notification` queue retries five times with backoff, skips deactivated users and uses a lease (a time-limited lock) to prevent double sending. Types include assignment, date change, cancellation, report ready, booking, agreement, overdue, contact update, SLA and warranty.

**Messages: Messages** collects customer conversations. Threads are scoped to a department or customer; office users see their own departments, portal users only their own threads. Outgoing messages can be sent as SMS with consent; incoming SMS lands in the same thread and is idempotent per provider ID.

## 11. API keys and the REST API

The API is under `/api/v1` and uses Bearer keys prefixed `fsm_`. NextAuth cookies are never accepted.

**Create a key:** go to **Settings** → **API keys**, enter **Name** and optional **Expires (optional)**, select **Create key**, copy the key into a secret manager and select **I have saved the key**.

The key is shown only once. The server stores only a SHA-256 hash and a short display prefix. If you lose it, revoke it and create a new one.

The API has no separate scopes. A key inherits exactly the owner's permissions, department and customer scope. Role and permission changes also apply to the key; a deactivated user gets `401`. **Important:** in the UI, the key is created for the signed-in administrator and therefore gets all administrator permissions. Treat it as an administrator secret. For lower privilege, create a dedicated active integration user with a narrow role and use `scripts/create-api-key.mjs` with `API_KEY_USER_EMAIL`, `API_KEY_NAME` and optionally `API_KEY_EXPIRES_AT`.

**Revoke** takes effect immediately and returns `401`. Expired keys also return `401`. The list shows **Owner**, **Created**, **Last used**, **Expires** and status **Active**, **Expired** or **Revoked**.

Each key can make **120 requests per minute**; exceeding the limit returns `429` with `Retry-After`. Pagination uses `page` and `limit` (maximum 100). OpenAPI 3.1 is at `/api/v1/openapi.json`; documentation with examples is in `docs/API.md`.

**Safe use:** use HTTPS, one key per integration, an expiry date, a secret manager and **Last used**. To rotate: create a new key, deploy it, verify with `GET /me`, then revoke the old one. On suspected leakage, revoke first, create a new key and review the activity log.

## 12. Activity log, privacy and data

**Activity log: Settings → Activity log** shows system events, security events such as failed 2FA attempts, and user activity with search, **From date** and **To date**, 50 rows per page. Work orders and customers have their own activity tabs; reports have revisions. The log is written in the same transaction as the mutation, cannot be edited or deleted in the UI, and is not hash-chained in 0.8.9.

**Privacy:** collect only necessary data, use department and customer scope, and grant `reports.viewAllDepartments` only when needed. Use separate portal accounts and deactivate them on leaving. Assess DPIA before tracking in a new department; tracking is opt-in, limited to `EN_ROUTE`, shows no route or coordinates and deletes pings after 24 hours. Obtain SMS consent and keep reviews internal. Deleting a customer deactivates portal users.

**Export:** there is no automatic export job or self-service GDPR export. Evidence is exported through the database backup or an administrator-controlled query against `WorkOrder`, `Report`, `ReportRevision` and `AuditLog` for one work order at a time. Never export reports from the client or a public route.

**Retention:** `RETENTION_DAYS` (default 365) controls hard deletion of soft-deleted work orders without reports. Work orders with reports are kept indefinitely. Legal hold stops all automatic deletion. Files are removed only when no rows reference them.

**Backup and upgrade:** take a backup before migration and rollback. Follow `docs/OPERATIONS.md`, use `scripts/backup.sh` and `scripts/restore-test.sh`. In production, set `REQUIRE_BACKUP_BEFORE_MIGRATE=true` so the app refuses to migrate without `BACKUP_CONFIRMED=true`. That only confirms a backup was taken, not that it is good; run the restore test first. Restore database and uploads from the same timestamp, stop the app before switching databases, use `prisma migrate deploy` and never `prisma db push`. Pin `APP_IMAGE` to a published version or digest, not `latest`, and never run `docker system prune --volumes`.

## 13. Operations and health

`GET /api/health` is liveness and returns 200 when the Node process is alive, without database, disk or queue calls; Docker uses it so a slow database does not cause a restart loop. `GET /api/ready` is readiness: unauthenticated calls get a short status and a database check, rate-limited per IP. With `READY_SECRET` or `CRON_SECRET`, the upload volume and job queue are also checked; it returns 503 when something is missing. Production Compose uses readiness but does not restart on `unhealthy`; a database outage gives 503 from the proxy. Both return `requestId`.

**Migrations:** the app runs `prisma migrate deploy` at startup. `DATABASE_MIGRATION_URL` can be used for a DDL role, `DATABASE_RUNTIME_URL` for the app. To run migrations separately, start with `SKIP_MIGRATIONS=true` and run them manually. Verify with `npx prisma migrate diff --from-config-datasource --to-schema prisma/schema.prisma --exit-code`.

**Jobs:** `RECURRING_SCHEDULER=true` runs service agreements and maintenance plans every six hours and at startup. `POST /api/cron/recurring` and `/api/cron/cleanup` require `Authorization: Bearer <CRON_SECRET>`. `PDF_JOB_WORKER` controls OCR/document jobs with at most two concurrent jobs per instance; report PDFs render in-process with a separate concurrency limit. `SMS_WORKER` is independent.

**Uploads and OCR:** files are stored in `UPLOAD_DIR` (default `./uploads`, `/app/uploads` in Docker). `/uploads/...` requires sign-in and checks the `Upload` row. Limits are 15 MB for images and 50 MB for

PDFs. A rolling 24-hour quota of 2 GiB per user and 10 GiB total storage per customer also applies. Adjust it with `UPLOAD_QUOTA_USER_MB` and `UPLOAD_QUOTA_CUSTOMER_MB`; the proxy still needs its own body limit (55 MB in the Caddy example). Over quota the user gets a clear error, and the upload is not written. OCR uses `TESSDATA_DIR` and `/tmp`; large PDFs can fill `/tmp`, and the job then fails without filling the root filesystem. Check `DocumentJob`, `PDF_JOB_WORKER` and disk.

**Environment variables:** the most important are `DATABASE_URL`, `NEXTAUTH_URL`, `APP_URL`, `NEXTAUTH_SECRET`, `SMTP_*`, `SMS_*`, `SSO_*`, `SCIM_*`, `PDF_JOB_WORKER`, `RECURRING_SCHEDULER`, `CRON_SECRET`, `READY_SECRET`, `RETENTION_DAYS`, `UPLOAD_*`, `TESSDATA_DIR`, `GEOCODE_*`, `ROUTING_*`, `TRACKING_AVERAGE_SPEED_KMH`, `TRUST_PROXY` and `BACKUP_*`. `.env` must be `chmod 600` and never shared; changes require a new deploy or restart.

**Before escalating:** is `/api/health` 200 and `/api/ready` 200/503? Is `app` `healthy`, and is `RestartCount` increasing? What do the logs say? Does Postgres respond? Is there free disk and a writable upload volume? Are `DocumentJob`, `Notification` or `SmsMessage` stuck? Include `requestId`, timestamp and error message, never secrets or personal data.

## 14. Administrator routines

**Onboard a department:** create it, enter contact details, set business hours and holidays, create a default SLA policy and optional price list, leave tracking off until DPIA is assessed, and create employees.

**Onboard a user:** create with the right role and department, send the invitation, let the user set a password, activate, ask for 2FA, and test that the user sees the right work orders and not other departments.

**Customer and portal user:** create the customer, add a contact with email, create the portal user under **More → Portal users**, send the invitation, activate and test the portal. Deactivate the test account afterwards.

**Publish a template:** build it, set the report name, unit requirement and HMS type, test on an internal work order, check the PDF, **A comment is required for these options** and conditional fields, then link it to agreements or work orders. Inform office and technicians.

**Launch an agreement and SLA:** choose or create a policy, check business hours and holidays, create any price list, create the agreement with the right interval and checklist, add lines and coverage, set renewal and notice, activate and check the next service date and SLA clock.

**Rotate an API key:** create a new key with a clear name, store it in the secret manager, deploy it, verify with `GET /api/v1/me`, revoke the old key, confirm it returns `401`, and update the documentation.

**Monthly review:** review administrators, employees, roles and departments. Review API keys, **Last used** and expiry. Review the **Activity log** for deletions, role changes, unlocks and legal hold. Check 2FA, portal users and unlocked reports. Run `scripts/restore-test.sh` and check disk, `/api/ready`, `RestartCount` and pending migrations.

## 15. Common problems

| Symptom | Cause | Action |
|---|---|---|
| User cannot sign in | Inactive, missing password or deactivated. | Activate, send a new invitation, or use **Forgot password**. |

| Symptom | Cause | Action |
|---|---|---|
| Customer portal is empty | Portal user inactive, customer deleted or contact missing. | Activate, restore and link a valid contact. |
| Template is missing, or changes do not apply | Wrong category/permission, or the report has a frozen template snapshot. | Check category and `templates.view`; create a new report for new fields. |
| Report cannot be unlocked | Legal hold or missing administrator. | Release the hold with a reason, or use an administrator. |
| SMS is not sent | `console`, missing consent, allowlist or cap. | Set the provider, record consent, check the cap. |
| SMS webhook rejects | `SMS_WEBHOOK_SECRET` missing or signature wrong. | Set the secret and check the signature. |
| Upload is rejected by quota | User or customer quota reached, or the proxy rejects the body. | Check `UPLOAD_QUOTA_*`, disk and the proxy body limit; remove unused attachments. |
| Unit cannot be moved | Active contract lines, number conflict or cross-customer reference. | Clear the contract lines, choose another order number, or check the preview. |
| SLA has the wrong deadline | Wrong policy, business hours or holiday. | Check the policy, department and calendar. |
| API returns `401` / `403` / `429` | Expired key, missing permission or rate limit. | Create a new key, grant permission, or wait for `Retry-After`. |
| `/api/ready` returns 503 | Database, upload volume or job queue. | Check Postgres, `UPLOAD_DIR`, disk and queue. |
| App restarts in a loop | Migration gate, OOM or `.env` error. | Check logs, `dmesg` and `REQUIRE_BACKUP_BEFORE_MIGRATE`. |

## 16. Relationship to the office and technician chapters

The administrator chapter covers setup and governance; the office and technician chapters cover daily use. **Department and role** determine which work orders and actions staff see; the customer portal is controlled by the customer link. **Templates, SLA, branding and SMS** provide the forms, deadlines and messages staff work with. **Legal hold** freezes data for all parties.

See `10-office.md` for customer cards, work orders, agreements, the booking queue and reports. See `20-technician.md` for the field app, offline drafts, forms, submission and tracking. See `90-reference.md` for concepts, error codes and reference material.

# Reference, glossary and troubleshooting

*Applies to FSM Platform version 0.8.9, 4 November 2026.*

This chapter is the reference section. It explains permissions, statuses and terms across roles, shows how features connect, and helps you find where to look when something is missing.

## Roles and permissions in practice

Permissions have more levels than "yes" and "no". **Full** means view, create, edit and administer. **Own department** means the role acts within its own department for areas that are department-scoped. **Own customer** means the portal user only sees their own customer. **No** means the area is not available.

| Area | Administrator | Office | Technician | Customer portal user |
|---|---|---|---|---|
| **Work orders and visits** | Full, all departments | View, create, edit, assign and change status in own department | View own department, create their own work orders, claim available work orders, change status and add/remove people | View their own visits |
| **Customers and contacts** | Full | Administer; customers are shared | View | View their own customer and contacts |
| **Equipment and meter readings** | Full | Administer; equipment is shared | View | View their own equipment if delegated |
| **Service contracts** | Full | View and administer in own department | View in own department | View their own contracts if delegated |
| **Administer SLA and business hours** | Full | No | No | No |
| **Bookings** | Full | Confirm, reschedule and cancel in own department | View in own department | Book and cancel their own if delegated |
| **Reports and checklists** | Full, including unlocking and legal hold | View and complete in own department; administer templates | View and fill in reports in own department | View submitted reports and acknowledge if they have document access |

| Area | Administrator | Office | Technician | Customer portal user |
|---|---|---|---|---|
| **Documents** | Full | Administer; documents are shared | View | View their own shared documents if delegated |
| **Employees and roles** | Full | Administer users in own department; can assign roles whose permissions are a subset of their own, never **Administrator**; role definitions and department moves are administrator-only | No | No |
| **Settings and branding** | Full | Administer branding, map, templates and dispatch; departments, roles and API are administrator-only | No | No |
| **Audit log** | Global | Per work order and customer | Per work order | No |
| **API keys** | Full | No | No | No |
| **Deletion, restore, unlocking and legal hold** | Full | No | No | No |

> **Note:** An administrator can create custom roles and give individual users tailored permissions. The table shows the standard roles. Check the user's actual permissions in "Settings" → "Employees & Roles" if something looks wrong.

Deletion is deliberately not a normal permission. An administrator can delete customers, files and other records. Work orders can be deleted by an administrator, or by a custom role explicitly granted `workorders.delete`, and only while the work order is open (not completed or cancelled). Reports and attachments are kept on soft deletion. Many deletions are soft, so the record can be restored.

# Status glossary

## Work order status

| Status | Displayed as | Meaning | Set by |
|---|---|---|---|
| `OPEN` | Open / New | Created but not planned. Used for bookings without an available slot, among other things. | System, office |
| `PENDING` | Waiting to be planned | Starting status for new and cloned work orders. Not selectable manually. | System |

| Status | Displayed as | Meaning | Set by |
|---|---|---|---|
| `PLANNED` | Planned | Has a date or visit and is ready for assignment or execution. | Office, system |
| `WAITING` | Waiting for customer | Work is on hold for the customer, access or clarification. | Office, technician |
| `IN_PROGRESS` | In progress | Work has started. | Technician, office |
| `COMPLETED` | Completed | Closed and treated as history. | Technician, office |
| `CANCELLED` | Cancelled | Closed without being carried out. | Office |

All open statuses can be changed freely, including to `COMPLETED` or `CANCELLED`. `COMPLETED` and `CANCELLED` are closed. Only an administrator can reopen a closed work order, and must give a reason.

## Visit status

| Status | Displayed as | Meaning |
|---|---|---|
| `PLANNED` | Planned | Planned, but no technician is assigned. |
| `ASSIGNED` | Assigned | At least one technician is assigned, but work has not started. |
| `EN_ROUTE` | On the way | The technician has started driving. Records the first response in any SLA. |
| `ON_SITE` | On site | The technician has arrived and is working on site. |
| `DONE` | Completed | The visit is finished. Tracking links are revoked. |
| `CANCELLED` | Cancelled | Cancelled before completion. |

The flow is `PLANNED → ASSIGNED → EN_ROUTE → ON_SITE → DONE`. `CANCELLED` can be used before the visit is completed. `DONE` and `CANCELLED` are closed.

> **Tip:** "On the way" starts the SLA response clock and makes it possible to share position and give the customer an estimated arrival.

## Report, unit and acknowledgement

| Term | Meaning |
|---|---|
| **Draft** | Completed fields saved locally or as a `ReportDraft`, but not submitted. A draft is not a report and is not shown to the customer. |
| **Submitted and locked** | The form has been submitted. The report is locked, has a PDF and a revision, and cannot be changed without an administrator. |

| Term | Meaning |
|---|---|
| **Open for correction** | An administrator has unlocked it with a reason. Changes are saved as a new revision, and the report locks automatically again. |
| **Acknowledged** | The customer has confirmed receipt. The acknowledgement is visible to the office and can be withdrawn on resubmission or unlocking. |
| **Revision** | An immutable snapshot before a change. Preserves names, units, answers, signatures and acknowledgement. |

| Unit result | Displayed as | Meaning |
|---|---|---|
| `GODKJENT` | Approved / done | The unit has been checked and is in order. |
| `AVVIK` | Deviation found | The unit has a deviation that needs follow-up. |
| Not assessed | Not assessed | The unit has no result. The report cannot be submitted until every unit is assessed. |

The overall report result is derived from the units: at least one deviation gives an overall deviation, all approved gives approved, and otherwise "not assessed" is shown. If the report contains deviations, one automatic remediation work order is created.

## Service contract and SLA

| Contract status | Displayed as | Meaning |
|---|---|---|
| `DRAFT` | Draft | Being set up; does not generate work orders. |
| `ACTIVE` | Active | Valid and can generate work orders when the next service is due. |
| `PAUSED` | Paused | Paused; does not generate work orders. |
| `EXPIRED` | Expired | Passed its end date without renewal. |
| `TERMINATED` | Terminated | Ended before its planned end. |

The renewal mode is `AUTO` (renews automatically) or `MANUAL` (must be renewed by the office). `Next service` is the date the system uses to decide whether to create a new work order.

| SLA state | Displayed as | Meaning |
|---|---|---|
| `ON_TRACK` | On track | Both response and resolution deadlines are within limits. |
| `AT_RISK` | At risk | The deadline is approaching; a notification is sent on the transition. |
| `BREACHED` | Breached | The deadline has passed; a notification is sent on the transition. |
| `MET` | Met | Response and resolution were recorded within the deadline. |
| `CANCELLED` | Cancelled | The work order was cancelled and the clock stopped. |

SLA deadlines are calculated using the department's business hours and holidays. The first "On the way" or "On site" records the response. "Completed" on the visit or work order records the resolution. If there is no active SLA policy, no SLA is shown.

> **Note:** The system does not calculate prices or invoices. Price lists and contract amounts are reference data.

## Booking status

| Status | Displayed as | Meaning |
|---|---|---|
| `PENDING` | Pending | In the office's booking queue and not tied to an available slot. |
| `CONFIRMED` | Confirmed | The slot is reserved, and the work order and visit have been created. |
| `CANCELLED` | Cancelled | The booking was cancelled. |
| `EXPIRED` | Expired | A pending request expired without confirmation. |

New bookings are normally confirmed directly against an available slot. The slot is held for five minutes while the customer completes the form. Older or manually created requests may remain `PENDING`. The event log uses `CREATED`, `CONFIRMED`, `RESCHEDULED`, `CANCELLED` and `EXPIRED`.

## Notifications and audit log

A notification is a message to one user about an event. It appears in the bell and under "Notifications", and can also be sent by email. The user chooses which types appear in the app and which are sent by email.

| Notification type | Typical recipient | Triggered by |
|---|---|---|
| `ASSIGNMENT` | Technician | The work order is assigned |
| `DATE_CHANGED` | Technician | The work order date changes |
| `CANCELLED` | Technician | The work order is cancelled |
| `REPORT_READY` | Technician, office and customer portal | The report is submitted and ready |
| `BOOKING` | Office and assigned technician | A new or confirmed booking |
| `CONTRACT_DUE` | Office | A service contract is due |
| `OVERDUE` | Technician and office | The work order is overdue |
| `CONTACT_UPDATE` | Office | The customer requests updated contact information |
| `SLA_AT_RISK` / `SLA_BREACHED` | Technician and office | The SLA deadline is approaching or has been breached |
| `WARRANTY` | Office | Equipment warranty is about to expire |

| Notification type | Typical recipient | Triggered by |
|---|---|---|
| `MESSAGE` | Office or customer | A new message in a conversation |

The audit log is a permanent, non-editable history of actions: who created, changed, moved, cancelled, deleted, unlocked or applied a legal hold, with time and metadata. You find it on the work order, on the customer card and under "Settings" → "Activity log" for administrators.

## How features talk together

> **Note:** `WorkOrder`, `VisitTechnician`, `BookingRequest` and `MaintenancePlanOccurrence` are internal model names. In the interface they are called work order, technician assignment, booking and maintenance occurrence.

| Action | Data created or updated | Who sees it |
|---|---|---|
| Customer books an available slot | `BookingRequest(CONFIRMED)`, `WorkOrder(PLANNED)`, `Visit(PLANNED)`, booking event, `BOOKING` | Customer, office, technician |
| Customer sends a request without a slot | `BookingRequest(PENDING)`, `WorkOrder(OPEN)`, booking event | Office in "Bookings" |
| Office confirms a pending booking | `BookingRequest(CONFIRMED)`, `WorkOrder(PLANNED)`, `Visit(PLANNED)`, optional SMS | Customer, office, technician |
| Office creates a work order | `WorkOrder(PENDING)`, `Visit(PLANNED)` if dated, audit log | Office, calendar |
| Office assigns a technician | `VisitTechnician`, technician list, `ASSIGNMENT`, audit log | Technician, office |
| Technician starts driving | `Visit(EN_ROUTE)`, `enRouteAt`, SLA response, tracking | Office, customer |
| Technician marks "On site" | `Visit(ON_SITE)`, `onSiteAt` | Office, customer |
| Technician completes the visit | `Visit(DONE)`, `doneAt`, tracking revoked | Office, customer |
| Technician submits the report | `Report` locked, revision 0, PDF, units, equipment history, `REPORT_READY`, optional remediation work order | Office, customer, equipment history |
| Customer acknowledges | `customerAcknowledgedAt`, acknowledgement comment | Office, report |
| Service contract becomes due | `WorkOrder(PLANNED)`, `Visit`, previous report as PDF, `CONTRACT_DUE` | Office, calendar |
| A meter reading passes the threshold | `MaintenancePlanOccurrence`, new maintenance work order | Office, customer |

| Action | Data created or updated | Who sees it |
|---|---|---|
| A document is uploaded | `Document`, `Upload`, OCR job, search index | Staff with document access |
| Role or permissions change | `User.sessionVersion` increases, API keys inherit new permissions | User, administrator |

> **Tip:** When troubleshooting, start with the action and follow the row in the table. You will see which side is missing data, and which role should see it.

## Common cross-role problems

| Problem | Check first | Where to look |
|---|---|---|
| **The customer portal user cannot sign in or see data** | Is the user active and linked to a contact belonging to the right customer? Is the delegation active with the right sub-permissions? Has the invitation been completed and a password set? | "Customers" → customer card → **More** → **Portal users**; "Settings" → "Employees & Roles" |
| **The customer cannot see documents, contracts or equipment** | Does the contact have `canViewDocuments`, `canViewContracts` or `canViewEquipment`? Is the document owned by the customer and shared? Is the contract in the right department? | Customer card → **More** → **Portal users** and the portal's "Documents", "Contracts" and "Equipment" tabs |
| **The work order is missing from the calendar** | Does the work order have a date and a visit? Undated work orders have no visit. Is the visit cancelled or completed? Is the date outside the ±6-month window? Is a filter hiding it? | "Work orders" → visit section; "Calendar"; "Dispatch" |
| **The report is not visible** | Was the form submitted, or is it only a draft? Is the report locked and submitted? Do the date, customer and department filters match? Is it deleted? Has the PDF been generated? | "Work orders" → "Checklists"; "Reports"; customer portal → "Reports" |
| **The service contract is not generating a work order** | Is the contract `ACTIVE`, not `PAUSED` or `EXPIRED`? Is `Next service` due? Has the end date passed? Is the customer deleted? Has the work order already been generated? Is the scheduled job running? | "Contracts" → contract; customer card; operations log for recurring jobs |
| **The notification does not arrive** | Is the type enabled for the app or email? Is the user active and actually a recipient? Is the email in spam? Was the notification deduplicated? Is the email queue being processed? | "Notifications" → "Settings"; "Notifications"; operations log for email |
| **The offline draft is not syncing** | Are you on the same device, in the same browser and signed in as the same user? Has the app been reopened with a connection? Is the draft in the outbox? Is it older than seven days, too large, or rejected because the report is locked? | Field surface → offline banner and draft list; browser storage |

| Problem | Check first | Where to look |
|---|---|---|
| **The technician cannot see the work order** | Is the work order in the technician's department? Is it assigned, or available? Is it completed or cancelled? Is the date correct? Does the technician have "Take work order" access? | "Today"; "Work orders"; "Calendar" |
| **An action is rejected with "no access"** | Does the user have the right role and permission? Is the user in the right department? Is the account active? Is the work order locked or under legal hold? | "Settings" → "Employees & Roles"; the work order's audit log |
| **Sites or units from the wrong customer appear** | Wrong customer selected, or the page was not reloaded after switching customer | Switch customer again, reload the page and check the saved selection |
| **A unit cannot be moved** | Active contract lines, number conflict, cross-customer reference or invalid target | Read the blocker in the preview, clear the contract line, or choose the right customer/site |
| **An upload is rejected** | File too large, user/customer quota reached, or proxy limit | Check file size, `UPLOAD_QUOTA_*`, disk and the proxy body limit |
| **Customer transfer does not start** | 10 starts per hour used up, an existing job is reused, or the preview expired | Wait, follow the existing job, or upload and preview the package again |

## Customer portal: access is more than signing in

A portal user is always linked to a contact at a customer. The contact can have a delegation with separate choices for booking, documents, contracts and equipment. An inactive delegation closes the portal completely for that contact. A contact without a delegation row has full portal access for backward compatibility.

Check therefore that the user is active and belongs to the right customer, that the contact belongs to the same customer, that the delegation is active, and that the sub-permission for the missing area is enabled.

> **Note:** In 0.8.9 there is no dedicated form in the interface for editing the delegation. Restricted portal access must be configured and tested with the actual account.

## Calendar: date, visit and window

The calendar shows visits, not work orders themselves. A work order without a date has no visit and cannot appear. The calendar loads a limited window around the period you are viewing, and the list may be truncated. If a work order is missing, it is almost always because the work order is undated, the visit is deleted, cancelled or completed, the date is outside the window, a filter hides it, or it belongs to another department.

## Reports: draft, submitted and published

A report does not become visible to the customer just because the technician has started filling it in. The sequence is: draft → submit → the server locks and generates a PDF → the report becomes available under

"Reports" and in the customer portal → the customer can acknowledge it. If the report is missing, find out where in this chain it stopped. A draft that only exists locally is not a report.

### Service contracts: only active and due contracts generate work orders

A service contract does not generate work orders just because it exists. It must be `ACTIVE`, `Next service` must be due, and the contract must not have passed its end date. The system also does not create a new work order for the same contract and date twice. If the contract is `PAUSED`, `EXPIRED` or `TERMINATED`, the office must activate or renew it first.

### Notifications: app, email and recipients

Notifications are both in-app and email. A missing notification in the inbox does not necessarily mean it was not created. Check whether the user has disabled the type, whether the user is active and actually a recipient, whether the notification is under "Notifications" without an email, whether the email is delayed or in spam, and whether the same event was already notified with a deduplication key.

### Offline: a draft is not the same as sent

The field surface saves drafts locally when the connection disappears. The draft is sent when the app is open and the connection returns. It cannot be sent if the report is locked or under legal hold, if the draft is older than seven days or larger than the limit, if the device has lost its storage, or if a different user or browser opens the app. Submission requires a connection; signature is optional.

## Norwegian/English glossary

| Norwegian | English | Meaning |
|---|---|---|
| Avdeling | Department | Organisational unit that scopes access |
| Oppdrag | Work order | The service job that is planned and followed up |
| Visitt | Visit | Time block within a work order, with technicians |
| Montør | Technician | The person carrying out field work |
| Kontor | Office | The planning and administration role |
| Kunde | Customer | The company receiving service |
| Kontaktperson | Contact | The person at the customer |
| Sted | Site | Physical address or facility |
| Utstyr / enhet | Equipment / unit | The machine or system being serviced |
| Rapportenhet | Report unit | The unit as it appears in a report |
| Sjekkliste | Checklist | The points being checked |
| Mal | Template | Reusable checklist or HMS form |
| Rapport | Report | Completed and submitted checklist |

| Norwegian | English | Meaning |
|---|---|---|
| Avvik | Deviation | A finding that needs follow-up |
| Godkjent | Approved | The unit has been checked and is in order |
| Revisjon | Revision | Immutable snapshot before a change |
| Kvittering | Acknowledgement | Customer confirmation of receipt |
| Låst | Locked | Cannot be changed without an administrator |
| Serviceavtale | Service contract | Agreement for recurring service |
| Fornyelse | Renewal | Automatic or manual extension |
| SLA | Service level agreement | Agreed response and resolution time |
| Åpningstider | Business hours | The hours within which SLA deadlines are calculated |
| Bestilling | Booking | The customer's request or confirmed appointment |
| Ledig time | Available slot | A slot that can be booked |
| Reservasjon | Hold | Short-term reservation of a slot |
| Varsel | Notification | Message about an event |
| Aktivitetslogg | Audit log | Traceable history of actions |
| Rettighet | Permission | What a role or user can do |
| Bevaringsplikt | Legal hold | Freezes data against deletion and change |
| Dokumentbibliotek | Document library | Shared file area with folders and tags |
| HMS | HMS | Health, safety and environment |
| Måler | Meter | Equipment reading, for example hours |
| Vedlikeholdsplan | Maintenance plan | Calendar- or meter-based maintenance |
| Sporing | Tracking | Temporary sharing of status and arrival |
| Omtale | Review | Internal customer feedback after a visit |
| Kundeportal | Customer portal | The customer's own surface |
| Kladd | Draft | Locally saved, unsubmitted form |
| Kundeadskillelse | Customer scoping | Site and unit choices are limited to the selected customer |
| Enhetsflytting | Equipment move | Moving a unit with its service history to a new customer |
| Kundeoverføring | Customer transfer | The administrator's export/import of a customer package |

| Norwegian | English | Meaning |
|---|---|---|
| Kontorinnboks | Office inbox | Collected SMS conversations for the office |

## Support information to collect

When you report a problem, it helps to collect the information below. Write briefly what you did, what you expected, and what actually happened.

- **Version:** FSM Platform 0.8.9, and where you saw the problem (office surface, field surface, customer portal, PDF or API).
- **Time:** Date and time with time zone, preferably Europe/Oslo.
- **User:** Email, role, department and any customer. Never provide passwords or one-time codes.
- **Case:** Work order number or ID, title, customer and date. For a report: report number or ID. For a booking: reference. For a contract: contract number.
- **Environment:** Browser and version, operating system, phone or PC, and whether the field surface is installed as an app.
- **Network:** Were you online or offline? Did it happen in a draft, during submission or afterwards?
- **Error message:** Reproduce the text exactly, and take a screenshot showing the time, user and error message.
- **Reproduction:** Write down the steps you took, and whether the problem happens every time or only sometimes.
- **Notification:** For notification problems, provide the type, recipient, app or email, and whether the notification exists under "Notifications".
- **API:** Provide the key prefix (never the full key), endpoint, time, HTTP status, `error.code` and any `requestId`.
- **PDF and email:** State whether it concerns generation, display, download or sending, and whether it concerns a report or a contract.
- **Offline:** State whether the draft is in the outbox, whether it is older than seven days, and whether the same user and device were used.

> **Warning:** Never send passwords, one-time codes, full API keys or complete personal data in a support case. Mask screenshots where possible.

# About this manual

This manual describes FSM Platform 0.8.9 as of 2026-11-04. It is divided into an introduction, one chapter for each role and a reference chapter.

| | |
|---|---|
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The source is split by role so a single chapter can be updated without rewriting the whole document. Build the PDF with `node scripts/build-manual.mjs`. The generated PDF is a snapshot of the product configuration described in this edition; local settings, permissions and integrations may differ.